

a* (carotenoid proxy) under metal stress: stratified analyses

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a* (carotenoid proxy) under metal stress: stratified analyses

Generated 2026-10-08, rerun on the curated strain table (PR #4, #5). All numbers come from `report/tables/*.csv`. Scripts are in `scripts/` and `scripts/strat/`. Data-quality caveats are in `analysis/heavy_metal_data_problems/HEAVY_METAL_D`.

Question. Strains grow on plates with a heavy metal at increasing doses. Does CIELAB a* (red-green axis, used here as a carotenoid proxy) rise because of stress? Or do strains differ inherently in a*? Colony size confounds the answer. Every dose effect is therefore shown with and without size.

What changed in this version

- **Strain groups come from the curated database** (`strain_curation`, decisions D-45 to D-47 and D-53). The group “*R. mucilaginosa*” now means pure haploid strains only (173 strains in Cr and Cu, 172 in Pb). Hybrid diploids (33 strains) and *R. aff. mucilaginosa* (7 strains) are separate groups. Five strains were renamed *R. frigidialcoholis* (D-46) and five previously unnamed strains received a species (D-45).
- **Iron and Zinc are not analysed.** Their source tables are incomplete (D-56): Iron has 62 of 120 plates and Zinc 9 of 119 plates in the ingested files. This report covers Chromium, Copper and Lead only. Old Fe and Zn results from earlier versions of this report are withdrawn.
- **No population analyses.** The old population labels came from the Y-2510 reference work and are dropped (D-51, D-57). New DH4148 SNP-cluster labels do not exist yet.
- **Species tests include hybrids and *aff. mucilaginosa* as groups.** Ploidy and species are confounded in the hybrid group, so a hybrid effect is a combined effect.

Main dataset (used everywhere unless stated)

- **Wells.** One row per replicate well. The largest object in each well and image is kept.

- **Area cutoff: colony area of at least 2,000 px.** Well-images whose largest object is smaller are dropped (section 10 gives the evidence).
- **Time window.** Median over the last 24 h up to a metal-specific end point T, counted from the first image of each plate. T is the median plate imaging span: Cr 114 h, Cu 114 h, Pb 107.5 h.
- **Wells kept.** Wells need at least 2 images in the window. Cr 7,287 wells (307 strains), Cu 7,484 (319), Pb 4,606 (307). Share of unfiltered wells kept: Chromium 90%, Copper 98%, Lead 68%.
- **Original unfiltered results** (no area cutoff) are in section 14, beside the main numbers.

Summary

1. **There is no general “stress raises a*”.** The effect depends on the metal. Over the full dose range at fixed colony size: Cr -8.5, Cu -13.4 and Pb -4.3 a* units (the Pb fit is singular).
2. **The dose-response is not linear.** Dose-as-factor models at fixed size show Cr a* rising at low doses (+2.2 and +3.1 at doses 0.2 and 0.4) and falling at the top (-9.8 at 1.2). Pb rises at doses 5-15 (+4.9, +8.0, +9.5) and is near 0 at doses 20-30 (-0.0, -0.3, -0.2). Cu is flat at dose 5 (+0.1) and then falls (-3.2 at dose 10, -12.4 at dose 30), even though median colony area is at least 0.76 of control up to dose 25.
3. **a* depends on colony area even without stress, so the main dataset drops colonies below 2,000 px.** In unstressed colonies median a* is 0.2-2.4 below 1,500 px, 2.6-2.9 at 1,500-2,000 px, 3.9-4.6 at 2,000-3,000 px and 19-21 at 33,000 px (section 10). The cutoff was chosen from the unstressed Cr and Cu curves and is assumed for Pb. It removes mostly top-dose wells, so **top-dose estimates describe larger survivors** (Cr: 311 strains at the top dose become 241; median top-dose area 1,729 becomes 3,281 px). Cu is unchanged by any cutoff (-13.3 to -13.4). Cr moves from -8.4 (no cutoff) to -9.3 (5,000 px). Pb is unstable (-6.1 with no cutoff, -4.3 at 2,000 px, -6.0 at 3,000 px, +0.2 at 5,000 px), because few Pb colonies at doses 20-30 pass any cutoff.
4. **Species and ploidy groups differ in baseline a*.** Against pure haploid *R. mucilaginosa*, hybrid diploids have lower baseline a* in all three metals (size-adjusted: Cr -2.9, Cu -2.0, Pb -2.5; corrected $p < 0.01$). *R. aff. mucilaginosa* is lower in Cr (-3.3) and Pb (-3.4) and borderline in Cu (-2.5, corrected $p = 0.075$). *R. diobovata* is lower in all three (-3.2, -2.8, -5.2). The omnibus species tests are significant in every metal (corrected $p \leq 0.002$).
5. **Hybrid diploids respond less strongly than pure haploids.** Dose slopes at fixed size: Cr -5.9 against -8.4, Cu -12.2 against -14.2, Pb -3.5 against -4.8 ($p < 0.001$, < 0.001 , 0.04 against the reference).
6. **Phylogenetic signal depends on the scope and the root.** With the tree rooted on the *Cystobasidium* + *Pseudomicrostroma* outgroup clade, Pagel’s lambda for baseline a* in all strains with a tree tip is 0.79 (Cr), 0.49 (Cu) and 0.74 (Pb). Inside pure haploid *R. mucilaginosa* (161-163 strains) lambda is 0.91 for Cr ($p = 0.008$) and 0 for Cu and Pb. So only Cr shows a signal inside the species.
7. **Regression to the mean inflates the baseline-vs-change pattern.** Selecting the high-baseline third of strains on half of the replicate wells and measuring the change on the other half gives a smaller gap than the naive analysis: Cu -1.5 against -5.0, Cr -3.5 against -5.1, Pb -4.2 against -5.5.
8. **Strain-specific change in a* is only partly repeatable.** Correlation of the change between replicate halves: Pb 0.56, Cr 0.49, Cu 0.11.
9. **Run (batch) is a small share of variance (0-3%). Only Cr shows clear heterogeneity of the dose effect between runs** ($I^2 = 0.77$, $Q\ p = 0.004$; Cu 0.49, $p = 0.12$; Pb 0.53, $p = 0.095$). The Cr effect weakens from -13.8 (run d000320) to -4.3 (run d000323). Each run is a different strain subset, so run and strain set cannot be separated.
10. **Colour and morphology (section 12).** b* moves much less than a*. The hue angle shifts from red toward yellow in all three metals (+1.0 to +3.8 SD, total effect). Solidity falls and eccentricity rises under stress, mainly in Cr and Pb. Cu b* jumps between dose 0 and dose 5 while a* and size barely change. This looks like a plate or media difference and is not interpreted.

1. Models (short)

- Source: DuckDB table `heavy_metal_measurement` (517,371 colony rows; 5 metals). Only Chromium, Copper and Lead are used here. Named strains only. a* is `ColorLab_a*GeoMedian` (the geometric-median CIELAB a* of the colony pixels); b* and L* are the matching `GeoMedian` columns.
- Groups: the `species` column of `results/strat/strain_table.csv`, built by `s0_inputs.py` from the `strain_info` view (curated species, `ploidy_status`, `clade_marker`).
- Models (lme4, REML). The overall models (Tables 1 and 2, Figures 1-3) use strain and plate random effects. Plate is unique within a run, so the overall models have no separate run term. Tables 1 and 2 differ: Table 1 has a strain intercept and a correlated strain dose slope; Table 2 (dose as a factor) has strain and plate intercepts only. The species, regime, size-bin and per-run models have a strain intercept, an uncorrelated strain dose slope, and run and plate intercepts (per-run models have no run term). The per-species strata in section 11 (Table 24) have strain and plate intercepts only. Models without a strain slope give standard errors that are too small when strains respond

differently, so Table 2 and Table 24 intervals are optimistic.

- `dose_s` is concentration divided by the maximum concentration. “At fixed size” adds $\ln(\text{area})$ centred on the control mean. The size term is one common slope per model.
- Dose units are not given in the source. Cr spans 0-1.2 and the other metals 0-30. Metals are never pooled.

2. Overall dose response

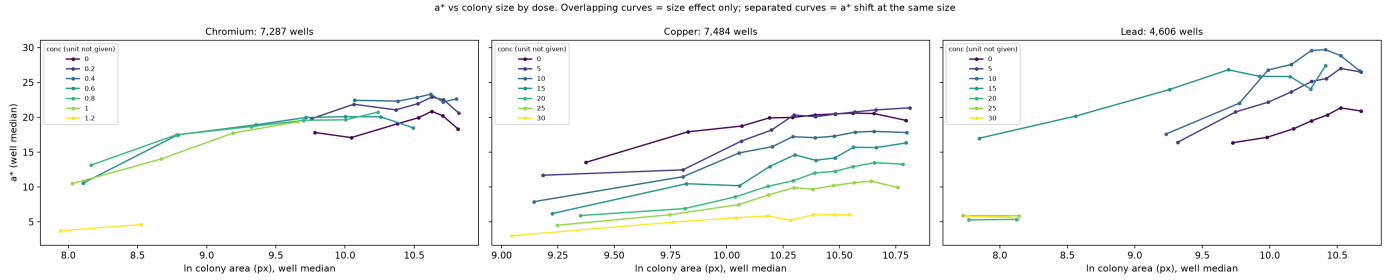


Figure 1. a^* against colony size by dose (well medians, binned). Overlapping curves mean a size effect only. Separated curves mean a shift at the same size.

Metal	model	n_wells	n_strains	dose effect (0 to top)	SE	p	strain share @0	strain share @top	singular
Chromium	total	7287	307	-12.65	1.05	<0.001	0.48	0.29	False
Chromium	at fixed size	7287	307	-8.54	0.98	<0.001	0.52	0.31	False
Copper	total	7484	319	-15.91	0.38	<0.001	0.50	0.38	False
Copper	at fixed size	7484	319	-13.39	0.31	<0.001	0.51	0.31	False
Lead	total	4606	307	-18.95	1.59	<0.001	0.38	0.05	True
Lead	at fixed size	4606	307	-4.31	1.23	<0.001	0.43	0.10	True

Table 1. Dose effect on a^* from 0 to the top dose. “total” has no size term. “at fixed size” adds $\ln(\text{area})$. “strain share” is strain variance / (strain + plate + residual) at dose 0 and at the top dose. Both Pb models are singular fits.

dose	Chromium	Copper	Lead
0.20	2.15		
0.40	3.11		
0.60	1.17		
0.80	0.92		
1.00	-2.93		
1.20	-9.83		
5.00		0.08	4.93
10.00		-3.21	7.96
15.00		-5.52	9.49
20.00		-8.65	-0.01
25.00		-10.57	-0.25
30.00		-12.36	-0.23

Table 2. a^* difference from 0 dose at fixed size (dose-as-factor model). Standard errors are 0.2-0.6 for Cr and Cu and 0.4-0.5 for Pb.

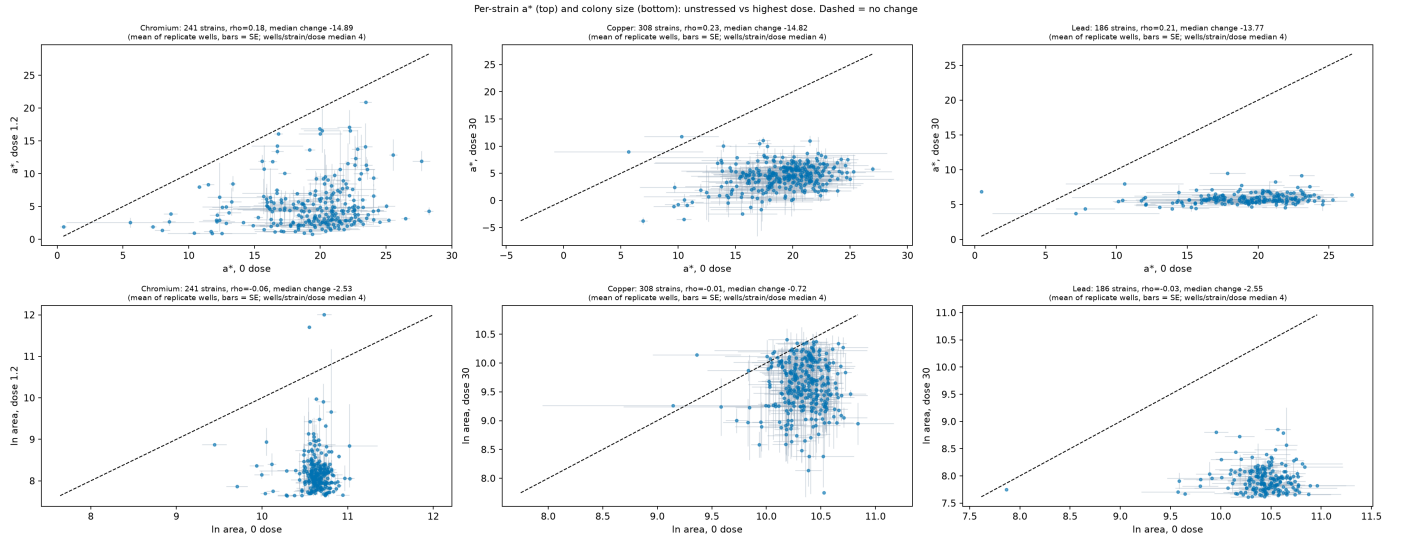


Figure 2. Per-strain a^* (top) and $\ln$ area (bottom), unstressed against top dose (mean of replicate wells, bars = SE).

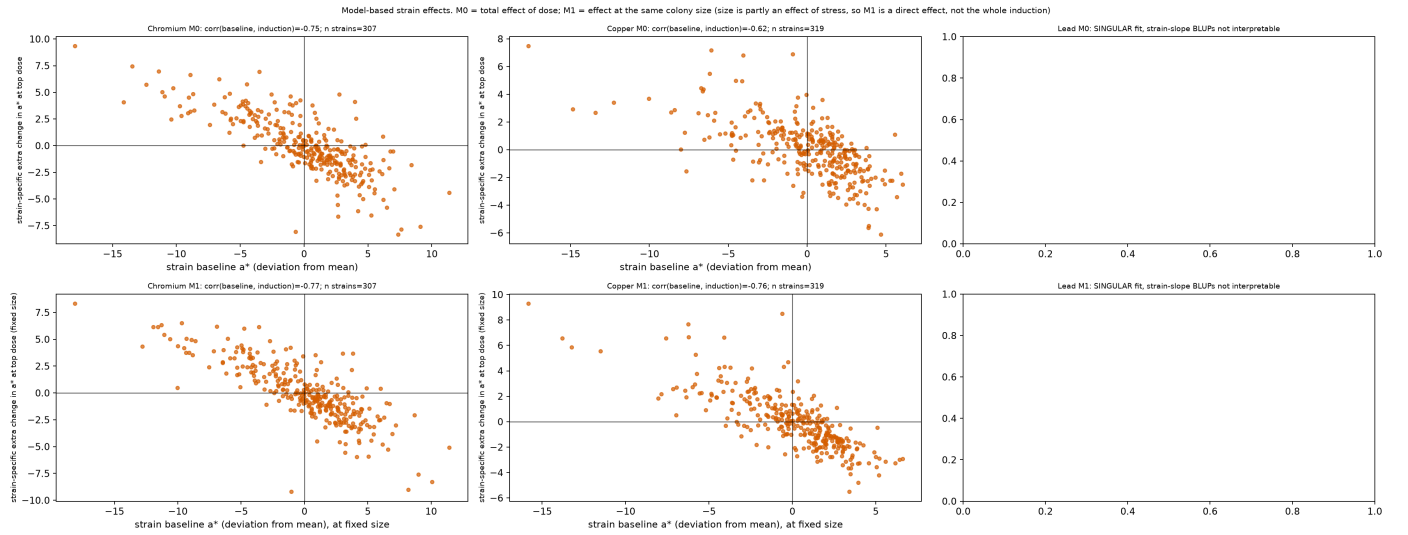


Figure 2b. Model-based strain effects: baseline a^* (x) against the strain-specific extra change in a^* at the top dose (y), total (top row) and at fixed size (bottom row). Singular fits are not drawn. Points that lie on a straight line (a correlation near -1 or $+1$) mean the random-effect correlation is at the model boundary. Those points are a rescaling of one random effect and carry no separate information.

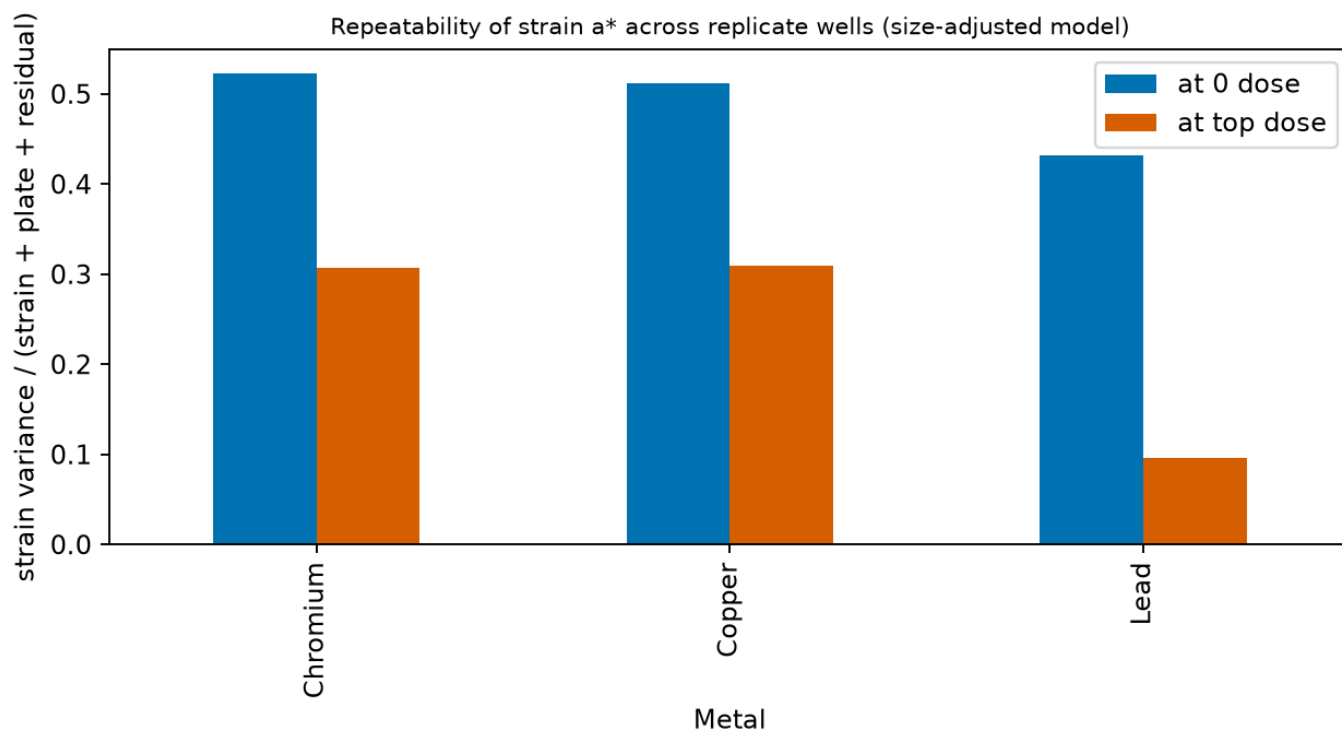


Figure 3. Share of a^* variance due to strain at 0 dose and at the top dose (size-adjusted model).

Reading. In all three metals strains converge toward a common low a^* at the top dose. The strain share falls from 0.52 to 0.31 (Cr), 0.51 to 0.31 (Cu) and 0.43 to 0.10 (Pb) (Table 1, size-adjusted model).

3. Which traits correlate with a^*

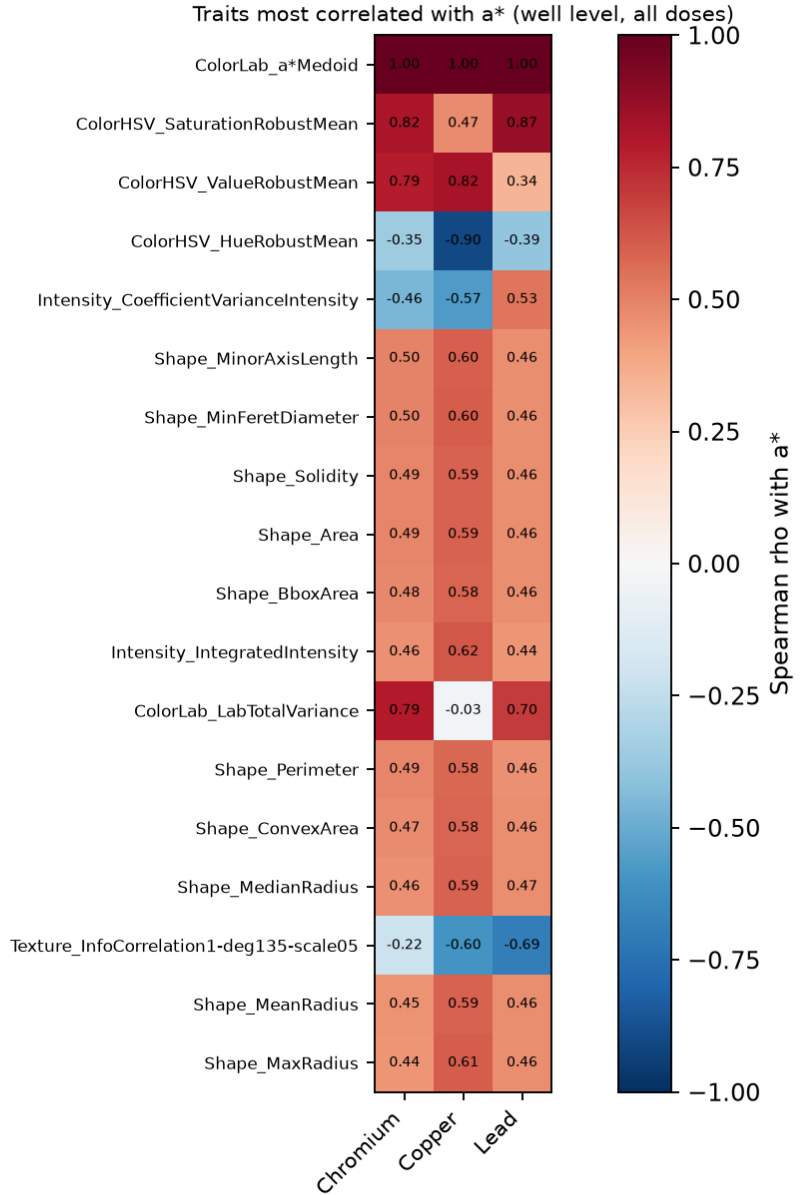


Figure 4. Spearman correlation of a^* with the traits most correlated with it (well level, all doses).

All doses (without ColorLab_a*Medoid, which is a^* itself):

trait	Chromium	Copper	Lead	mean abs rho
ColorHSV_SaturationRobustMean	0.82	0.47	0.87	0.72
ColorHSV_ValueRobustMean	0.79	0.82	0.34	0.65
ColorHSV_HueRobustMean	-0.35	-0.90	-0.39	0.55
Intensity_CoefficientVarianceIntensity	-0.46	-0.57	0.53	0.52
Shape_MinorAxisLength	0.50	0.60	0.46	0.52
Shape_MinFeretDiameter	0.50	0.60	0.46	0.52
Shape_Solidity	0.49	0.59	0.46	0.51
Shape_Area	0.49	0.59	0.46	0.51
Shape_BboxArea	0.48	0.58	0.46	0.51
Intensity_IntegratedIntensity	0.46	0.62	0.44	0.51

Control dose only:

trait	Chromium	Copper	Lead	mean abs rho
ColorHSV_SaturationRobustMean	0.85	0.61	0.89	0.79
ColorLab_LabTotalVariance	0.79	0.73	0.79	0.77
ColorHSV_HSVConcVariance	0.68	0.60	0.71	0.66
ColorHSV_ValueRobustMean	0.69	0.52	0.66	0.63
Intensity_StandardDeviationIntensity	-0.31	-0.62	-0.55	0.49
Texture_SumVariance-deg090-scale05	-0.21	-0.60	-0.52	0.44
Intensity_CoefficientVarianceIntensity	-0.23	-0.53	-0.56	0.44
Texture_HaralickVariance-deg090-scale05	-0.20	-0.59	-0.52	0.44
Texture_SumVariance-deg000-scale05	-0.21	-0.59	-0.51	0.44
Texture_HaralickVariance-deg000-scale05	-0.20	-0.59	-0.51	0.44

Within strain and dose (replicate wells only):

trait	Chromium	Copper	Lead	mean abs rho
ColorHSV_SaturationRobustMean	0.82	0.61	0.80	0.74
ColorHSV_ValueRobustMean	0.72	0.45	0.67	0.61
ColorLab_b*Medoid	0.64	0.40	0.50	0.51
ColorLab_b*GeoMedian	0.64	0.39	0.50	0.51
ColorLab_LabTotalVariance	0.66	0.20	0.63	0.50
Shape_MinorAxisLength	0.33	0.38	0.66	0.45
Shape_Solidity	0.32	0.38	0.66	0.45
Shape_MinFeretDiameter	0.32	0.38	0.66	0.45
Shape_ConvexArea	0.30	0.38	0.65	0.44
Shape_MedianRadius	0.28	0.38	0.66	0.44

Reading. Saturation, value and the colour-variance traits correlate most strongly with a*. These share colour information with a*. Size traits (area, radius, Feret diameter, integrated intensity) correlate at 0.44-0.62 in all three metals. Many size traits are near-duplicates, so treat them as one family. Replicate wells within a strain and dose still show an area-a* correlation of 0.28 (Cr), 0.38 (Cu) and 0.64 (Pb), so the link is not only between strains.

4. Species and ploidy groups (stratification item 1)

Group is a fixed effect (reference: pure haploid *R. mucilaginosa*). Strain, run and plate are random. Groups with at least 5 strains are included. “Hybrid diploid” and “aff. mucilaginosa” are groups defined by the curation table; the hybrid group differs from the reference in both species composition and ploidy.

Metal	model	F	p	p_BH
Chromium	baseline	8.05	<0.001	<0.001
Chromium	baseline_size_adjusted	8.05	<0.001	<0.001
Chromium	dose_response_size_adjusted	9.94	<0.001	<0.001
Copper	baseline	2.77	0.00	0.00
Copper	baseline_size_adjusted	2.98	0.00	0.00
Copper	dose_response_size_adjusted	3.14	<0.001	0.00
Lead	baseline	7.98	<0.001	<0.001
Lead	baseline_size_adjusted	8.07	<0.001	<0.001
Lead	dose_response_size_adjusted	4.75	<0.001	<0.001

Table 3. Omnibus tests (*F* test). *p*_{BH} is the Benjamini-Hochberg adjustment across all 9 rows (baseline, size-adjusted baseline and dose-response tests in three metals).

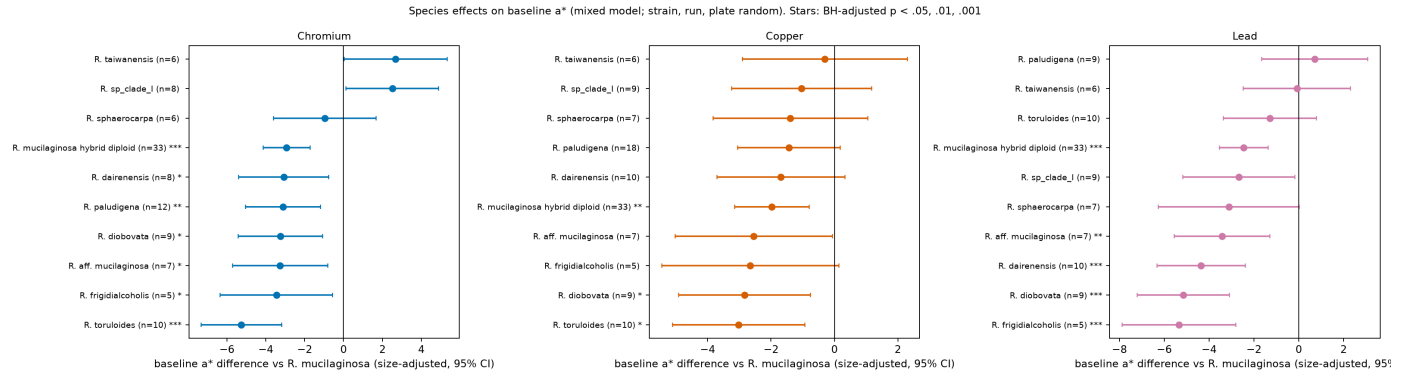
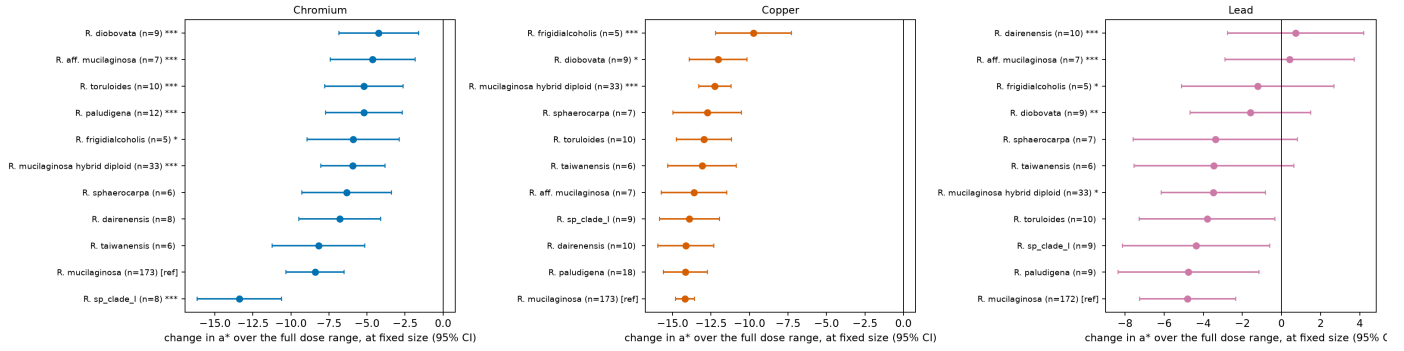


Figure 5. Group effect on baseline a^* relative to pure haploid *R. mucilaginosa* (size-adjusted).

Metal	species	n_strains	estimate	se	p	p_BH
Chromium	<i>R. aff. mucilaginosa</i>	7	-3.26	1.25	0.01	0.02
Chromium	<i>R. dairenensis</i>	8	-3.08	1.18	0.01	0.02
Chromium	<i>R. diobovata</i>	9	-3.25	1.11	0.00	0.01
Chromium	<i>R. frigidialcoholis</i>	5	-3.46	1.47	0.02	0.04
Chromium	<i>R. mucilaginosa</i> hybrid diploid	33	-2.92	0.62	<0.001	<0.001
Chromium	<i>R. paludigena</i>	12	-3.12	0.98	0.00	0.01
Chromium	<i>R. sp_clade_I</i>	8	2.51	1.22	0.04	0.07
Chromium	<i>R. sphaerocarpa</i>	6	-0.96	1.35	0.48	0.53
Chromium	<i>R. taiwanensis</i>	6	2.68	1.35	0.05	0.08
Chromium	<i>R. toruloides</i>	10	-5.25	1.05	<0.001	<0.001
Copper	<i>R. aff. mucilaginosa</i>	7	-2.55	1.26	0.05	0.08
Copper	<i>R. dairenensis</i>	10	-1.69	1.03	0.10	0.13
Copper	<i>R. diobovata</i>	9	-2.83	1.06	0.01	0.02
Copper	<i>R. frigidialcoholis</i>	5	-2.65	1.43	0.06	0.09
Copper	<i>R. mucilaginosa</i> hybrid diploid	33	-1.97	0.60	0.00	0.01
Copper	<i>R. paludigena</i>	18	-1.43	0.83	0.08	0.11
Copper	<i>R. sp_clade_I</i>	9	-1.03	1.13	0.36	0.42
Copper	<i>R. sphaerocarpa</i>	7	-1.39	1.25	0.27	0.32
Copper	<i>R. taiwanensis</i>	6	-0.30	1.33	0.82	0.85
Copper	<i>R. toruloides</i>	10	-3.02	1.07	0.00	0.01
Lead	<i>R. aff. mucilaginosa</i>	7	-3.42	1.09	0.00	0.01
Lead	<i>R. dairenensis</i>	10	-4.35	1.01	<0.001	<0.001
Lead	<i>R. diobovata</i>	9	-5.15	1.05	<0.001	<0.001
Lead	<i>R. frigidialcoholis</i>	5	-5.35	1.29	<0.001	<0.001
Lead	<i>R. mucilaginosa</i> hybrid diploid	33	-2.45	0.55	<0.001	<0.001
Lead	<i>R. paludigena</i>	9	0.71	1.21	0.56	0.59
Lead	<i>R. sp_clade_I</i>	9	-2.67	1.27	0.04	0.07
Lead	<i>R. sphaerocarpa</i>	7	-3.11	1.60	0.05	0.08
Lead	<i>R. taiwanensis</i>	6	-0.08	1.22	0.95	0.95
Lead	<i>R. toruloides</i>	10	-1.29	1.06	0.23	0.28

Table 4. Group baseline contrasts (size-adjusted).

Figure 6. Group-specific change in a^* over the full dose range at fixed size.

Metal	species	n_strains	slope_full_range	se	p vs mucilaginosa
Chromium	<i>R. mucilaginosa</i>	173	-8.41	0.98	
Chromium	<i>R. aff. mucilaginosa</i>	7	-4.63	1.42	<0.001
Chromium	<i>R. dairenensis</i>	8	-6.78	1.37	0.11
Chromium	<i>R. diobovata</i>	9	-4.23	1.33	<0.001
Chromium	<i>R. frigidialcoholis</i>	5	-5.91	1.54	0.04
Chromium	<i>R. mucilaginosa hybrid diploid</i>	33	-5.93	1.07	<0.001
Chromium	<i>R. paludigena</i>	12	-5.21	1.28	<0.001
Chromium	<i>R. sp_clade_I</i>	8	-13.39	1.41	<0.001
Chromium	<i>R. sphaerocarpa</i>	6	-6.34	1.50	0.08
Chromium	<i>R. taiwanensis</i>	6	-8.18	1.55	0.85
Chromium	<i>R. toruloides</i>	10	-5.21	1.31	<0.001
Copper	<i>R. mucilaginosa</i>	173	-14.19	0.32	
Copper	<i>R. aff. mucilaginosa</i>	7	-13.60	1.09	0.59
Copper	<i>R. dairenensis</i>	10	-14.15	0.93	0.96
Copper	<i>R. diobovata</i>	9	-12.05	0.96	0.02
Copper	<i>R. frigidialcoholis</i>	5	-9.74	1.26	<0.001
Copper	<i>R. mucilaginosa hybrid diploid</i>	33	-12.25	0.53	<0.001
Copper	<i>R. paludigena</i>	18	-14.17	0.73	0.98
Copper	<i>R. sp_clade_I</i>	9	-13.91	0.99	0.78
Copper	<i>R. sphaerocarpa</i>	7	-12.75	1.14	0.20
Copper	<i>R. taiwanensis</i>	6	-13.08	1.14	0.33
Copper	<i>R. toruloides</i>	10	-12.96	0.91	0.17
Lead	<i>R. mucilaginosa</i>	172	-4.80	1.25	
Lead	<i>R. aff. mucilaginosa</i>	7	0.41	1.69	<0.001
Lead	<i>R. dairenensis</i>	10	0.72	1.78	<0.001
Lead	<i>R. diobovata</i>	9	-1.60	1.57	0.00
Lead	<i>R. frigidialcoholis</i>	5	-1.21	1.99	0.02
Lead	<i>R. mucilaginosa hybrid diploid</i>	33	-3.49	1.36	0.04
Lead	<i>R. paludigena</i>	9	-4.75	1.84	0.97
Lead	<i>R. sp_clade_I</i>	9	-4.37	1.92	0.77
Lead	<i>R. sphaerocarpa</i>	7	-3.38	2.14	0.43
Lead	<i>R. taiwanensis</i>	6	-3.45	2.09	0.43
Lead	<i>R. toruloides</i>	10	-3.80	1.77	0.45

Table 5. Group-specific dose slopes (a^* change over the full dose range).

Metal	n_other_strains	baseline_diff	baseline_se	baseline_p	slope_diff	slope_se	slope_p
Chromium	104	-2.45	0.43	0.00	2.14	0.37	0.00
Copper	114	-1.91	0.38	0.00	1.29	0.33	0.00
Lead	105	-2.70	0.39	0.00	2.16	0.46	0.00

Table 6. Pure haploid *R. mucilaginosa* against all other groups pooled (contrast only; the pool includes hybrids and aff. *mucilaginosa*).

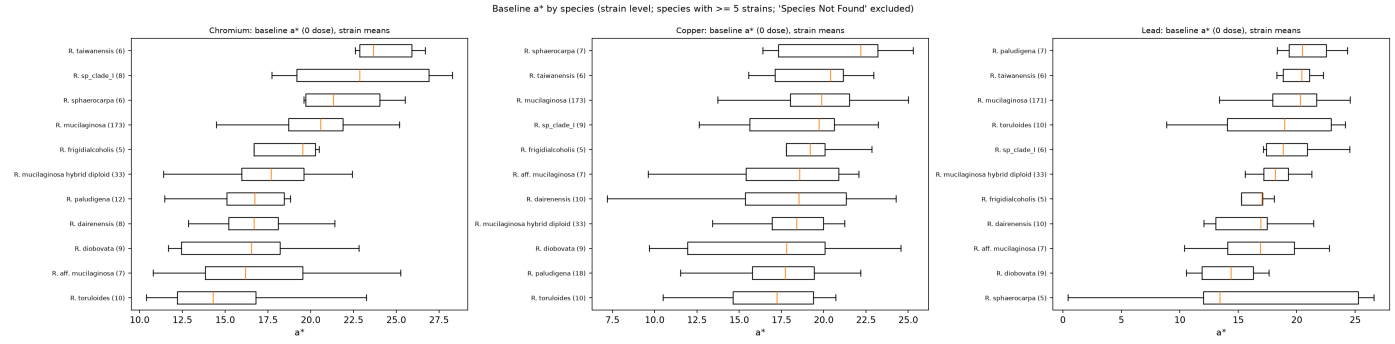


Figure 7. Baseline a^* by group (strain means; groups with at least 5 strains).

Reading.

- Hybrid diploids have lower baseline a^* than pure haploids in all three metals, and a weaker dose response in all three (Tables 4 and 5). The ploidy and the parental mixture cannot be separated here.
- Pooled other groups are lower in baseline a^* than pure haploid *R. mucilaginosa* in all three metals (-1.9 to -2.7) and respond less strongly (slope difference +1.3 to +2.2; Table 6).
- Species effects differ between metals. For example *R. paludigena* has no Pb baseline difference (+0.7) but is -3.1 in Cr. *R. sp_clade_I* is +2.5 in Cr and -2.7 in Pb. The non-mucilaginosa groups have 5-18 strains each, so single-group contrasts are noisy. Cu shows the weakest species signal (omnibus $F = 3.0$).
- R. sp_clade_I* responds most strongly in Cr (-13.4 over the full range against -8.4 for the reference). In Cu most groups match the reference (-12.0 to -14.2); *R. frigidicola* is weaker (-9.7).

5. Phylogenetic signal (item 2)

The PHYling tree is rooted on the outgroup clade made of its two non-Rhodotorula tips, *Cystobasidium* sp. DBVPG_10075 and *Pseudomicrostoma phylloplanum* DBVPG_6740. The root sits in the middle of the branch that separates them from the *Rhodotorula* strains. Lambda depends on the root, because the covariance under Brownian motion is built from the distance from the root to the common ancestor of each pair of strains.

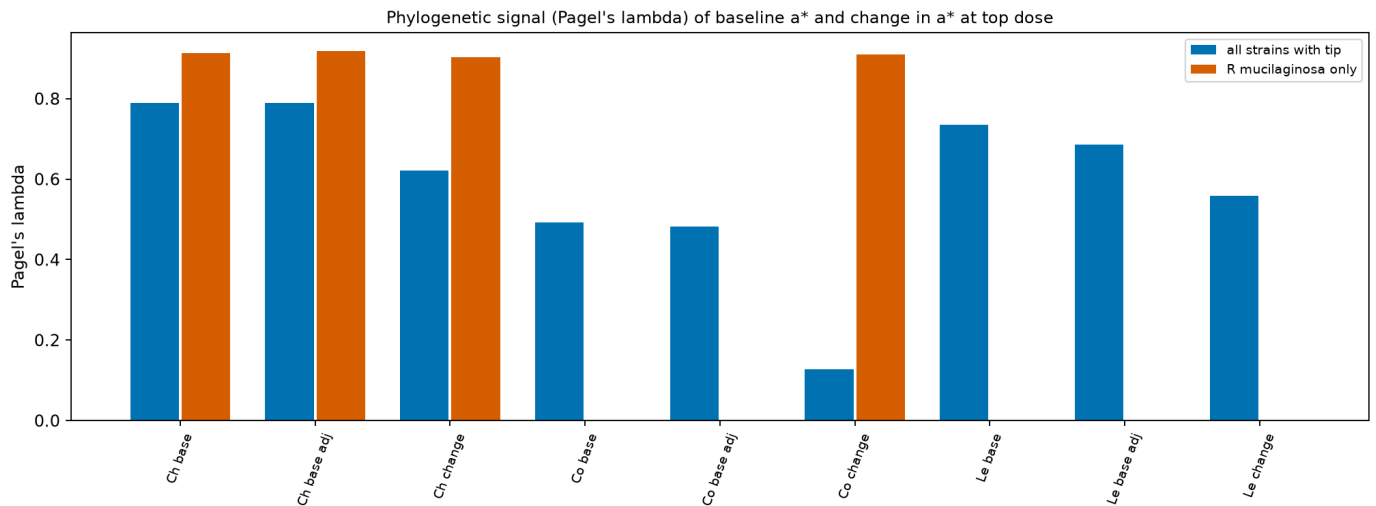


Figure 8. Pagel's lambda of baseline a^* and of the change in a^* at the top dose.

Metal	scope	trait	n_strains	pagel_lambda	p (lambda > 0)
Chromium	all_strains_with_tip	baseline a^*	255	0.79	<0.001
Chromium	all_strains_with_tip	baseline a^* (size-adjusted)	255	0.79	<0.001
Chromium	all_strains_with_tip	change in a^* at dose 1.2	209	0.62	<0.001
Chromium	<i>R. mucilaginosa</i> _only	baseline a^*	163	0.91	0.01

Metal	scope	trait	n_strains	pagel_lambda	p (lambda > 0)
Chromium	R_mucilaginosa_only	baseline a^* (size-adjusted)	163	0.92	0.01
Chromium	R_mucilaginosa_only	change in a^* at dose 1.2	135	0.90	0.01
Copper	all_strains_with_tip	baseline a^*	264	0.49	<0.001
Copper	all_strains_with_tip	baseline a^* (size-adjusted)	264	0.48	<0.001
Copper	all_strains_with_tip	change in a^* at dose 30	256	0.13	0.51
Copper	R_mucilaginosa_only	baseline a^*	163	<0.001	1.00
Copper	R_mucilaginosa_only	baseline a^* (size-adjusted)	163	<0.001	1.00
Copper	R_mucilaginosa_only	change in a^* at dose 30	163	0.91	0.19
Lead	all_strains_with_tip	baseline a^*	249	0.73	<0.001
Lead	all_strains_with_tip	baseline a^* (size-adjusted)	249	0.69	<0.001
Lead	all_strains_with_tip	change in a^* at dose 30	164	0.56	<0.001
Lead	R_mucilaginosa_only	baseline a^*	161	<0.001	1.00
Lead	R_mucilaginosa_only	baseline a^* (size-adjusted)	161	<0.001	1.00
Lead	R_mucilaginosa_only	change in a^* at dose 30	106	<0.001	1.00

Table 7. Pagel’s lambda (maximum likelihood under Brownian motion on the PHYling FastTree). 249-264 strains with a tree tip in the all-strains scope; 161-163 pure haploid *R. mucilaginosa* strains in the second scope.

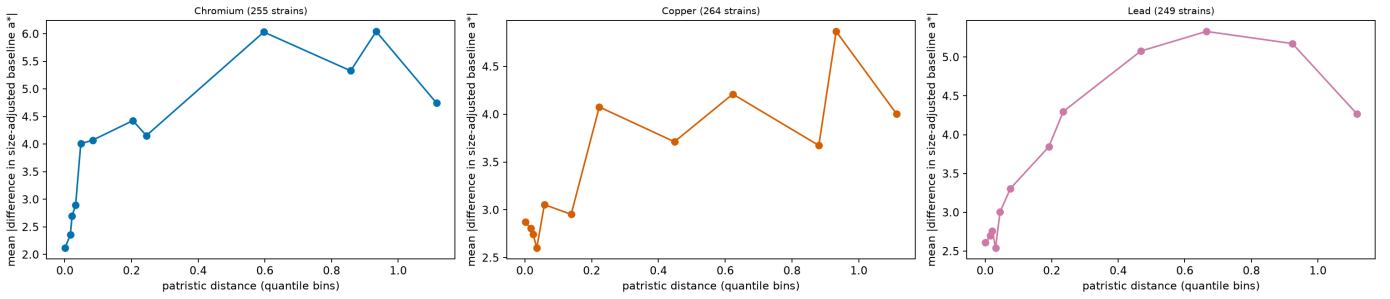


Figure 9. Mean absolute difference in size-adjusted baseline a^* between pairs of strains, by patristic distance.

PHYling tree (278 tips), rooted on the outgroup clade *Cystobasidium* + *Pseudomicrostroma*, with baseline a^* per metal

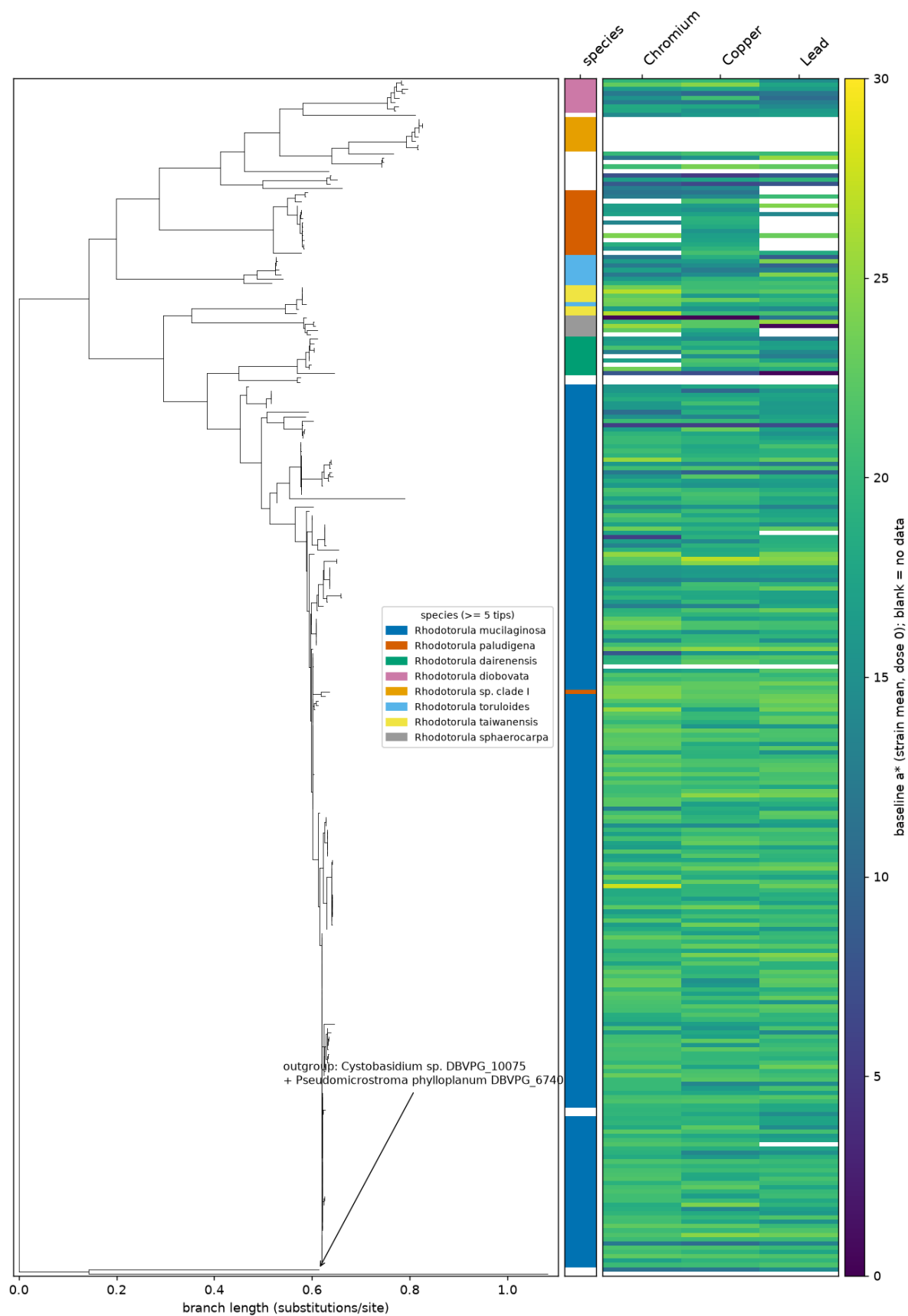


Figure 10. The tree rooted on the *Cystobasidium* + *Pseudomicrostroma* outgroup clade (bottom), with the species of each tip and baseline a^* per metal (strain mean at dose 0, one common colour scale; blank = no data).

Reading and limits.

- Closely related strains share baseline a^* . Lambda is 0.79 (Cr), 0.49 (Cu) and 0.74 (Pb) for all strains with a tip (Table 7). Inside pure haploid *R. mucilaginosa* it is 0.91 for Cr ($p = 0.008$) and 0.00 for Cu and Pb, so the signal in Cu and Pb is the separation between species and groups, not structure inside the haploid species. The pairwise difference in size-adjusted a^* increases with patristic distance (Figure 9; patristic distance does not depend on the root).
- **The root matters.** The first version of this report used the arbitrary root of the tree file and got different values. The outgroup-rooted values are used because the outgroup root is the biologically meaningful one.
- Lambda of the change in a^* at the top dose is 0.62 (Cr), 0.13 (Cu; $p = 0.51$) and 0.56 (Pb) for all strains. The change is top-dose a^* minus baseline a^* , and top-dose a^* sits on a floor with little variation (Pb SD 0.77 across strains), so for Pb, Cr and Cu the change partly restates the baseline.
- **Blomberg's K was dropped.** The tree has 22 zero-length tips and 74 tips with a neighbour closer than $1e-5$, so the covariance matrix is nearly singular (condition number about $2e10$ with the outgroup root). K varied from about 0 to 0.06 with the diagonal jitter (Table 8), so it is not interpretable. Lambda was stable across jitter $1e-8$ to $1e-3$ (for example 0.48 in Cu).
- Strains within species and groups are not independent (lambda 0.5-0.9 in the all-strains scope). The group tests in section 4 treat strain as the only grouping, so their p-values are too small.
- 266 of 321 strains have a unique tree tip. Strains without a tip are excluded.

Metal	jitter	cond	lam	K	pK
Copper	1e-08	1.98e+10	0.482	7.62e-07	0.005
Copper	1e-06	1.98e+08	0.482	3.82e-05	0.005
Copper	0.0001	1.98e+06	0.482	0.00147	0.005
Copper	0.001	1.98e+05	0.482	0.00952	0.005
Copper	0.01	1.98e+04	0.487	0.0585	0.015
Lead	1e-08	1.95e+10	0.686	8.72e-07	0.005
Lead	1e-06	1.95e+08	0.686	4.08e-05	0.005
Lead	0.0001	1.95e+06	0.686	0.00125	0.005
Lead	0.001	1.95e+05	0.688	0.00703	0.005
Lead	0.01	1.95e+04	0.696	0.0407	0.005
Chromium	1e-08	1.97e+10	0.789	8.03e-07	0.005
Chromium	1e-06	1.97e+08	0.789	4.88e-05	0.005
Chromium	0.0001	1.97e+06	0.789	0.00203	0.005
Chromium	0.001	1.97e+05	0.789	0.0132	0.005
Chromium	0.01	1.97e+04	0.799	0.0795	0.005

Table 8. Sensitivity of lambda and K to the diagonal jitter added to the covariance matrix (main dataset, outgroup-rooted tree).

6. Dose regimes (item 3)

A dose is **sub-inhibitory** when the median (across strains) of area at that dose / area at dose 0 is at least 0.5. Otherwise it is **inhibitory**. The 0.5 cutoff is a choice, not a measured threshold. The regime is defined from colony size, and size also relates to a^* , so the split is descriptive and partly circular.

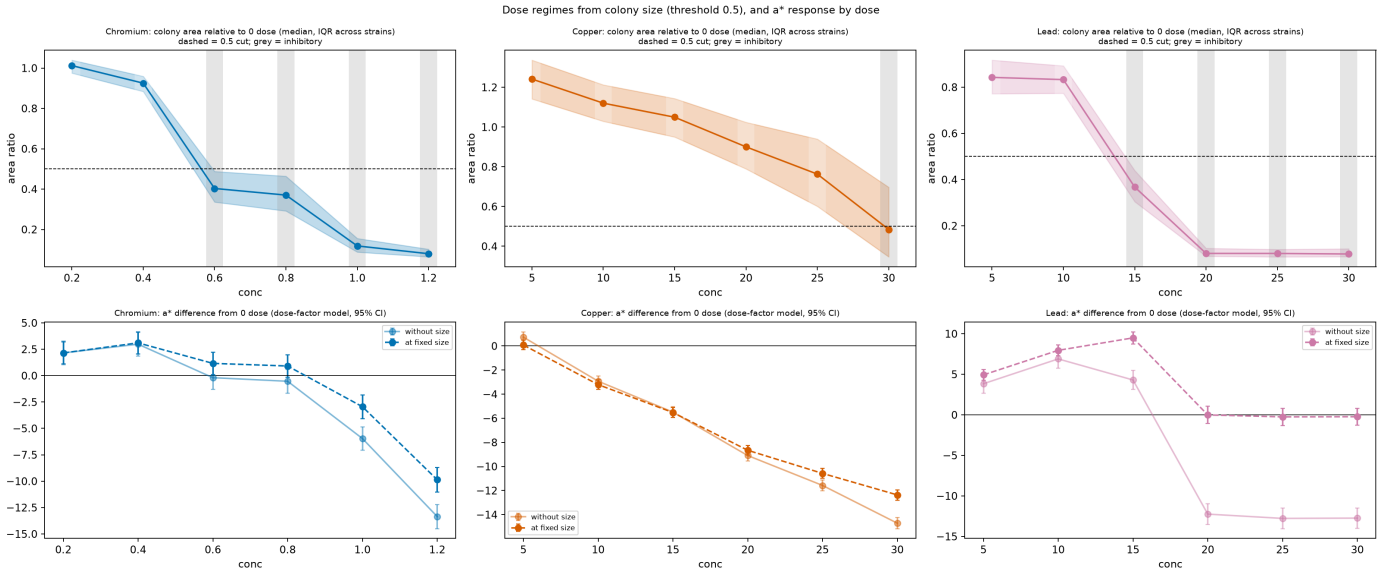


Figure 11. Top: area ratio by dose with the 0.5 cutoff. Bottom: a^* difference from 0 dose, with and without size.

Metal	conc	median_ratio	q25	q75	n_strains	regime
Chromium	0.20	1.01	0.98	1.04	305	sub-inhibitory
Chromium	0.40	0.93	0.88	0.96	305	sub-inhibitory
Chromium	0.60	0.40	0.34	0.49	305	inhibitory
Chromium	0.80	0.37	0.29	0.46	305	inhibitory
Chromium	1.00	0.12	0.09	0.16	292	inhibitory
Chromium	1.20	0.08	0.06	0.10	241	inhibitory
Copper	5.00	1.24	1.14	1.34	319	sub-inhibitory
Copper	10.00	1.12	1.03	1.21	318	sub-inhibitory
Copper	15.00	1.05	0.95	1.14	317	sub-inhibitory
Copper	20.00	0.90	0.79	1.02	313	sub-inhibitory
Copper	25.00	0.76	0.60	0.94	311	sub-inhibitory
Copper	30.00	0.48	0.35	0.70	308	inhibitory
Lead	5.00	0.84	0.77	0.92	295	sub-inhibitory
Lead	10.00	0.83	0.77	0.89	295	sub-inhibitory
Lead	15.00	0.37	0.30	0.44	288	inhibitory
Lead	20.00	0.08	0.07	0.10	165	inhibitory
Lead	25.00	0.08	0.07	0.10	177	inhibitory
Lead	30.00	0.08	0.07	0.10	186	inhibitory

Table 9. Colony area relative to 0 dose.

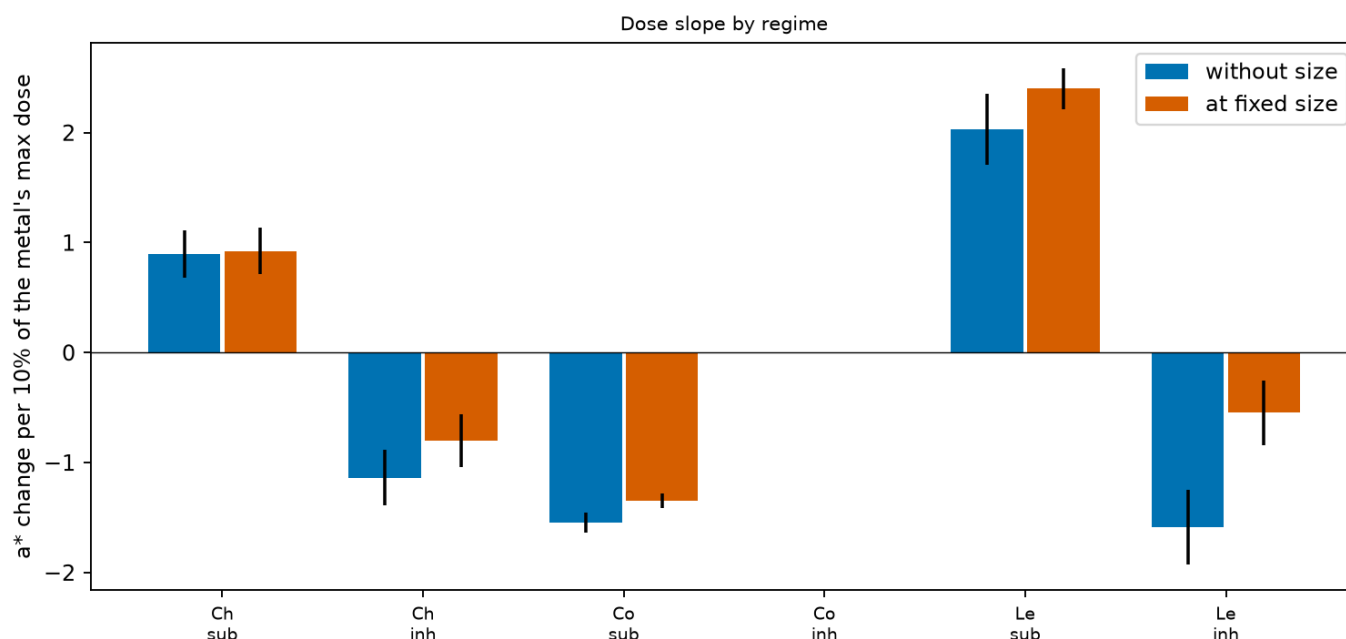


Figure 12. Slope of a^* per 10% of the maximum dose, by regime.

Metal	regime	doses	n_wells	size_adjusted	slope_per_0.1_of_max_dose	se	p
Chromium	sub-inhibitory	0,0,2,0.4	3562	False	0.89	0.11	<0.001
Chromium	sub-inhibitory	0,0,2,0.4	3562	True	0.92	0.11	<0.001
Chromium	inhibitory	0,0.6,0.8,1,1.2	4913	False	-1.14	0.13	<0.001
Chromium	inhibitory	0,0.6,0.8,1,1.2	4913	True	-0.80	0.12	<0.001
Copper	sub-inhibitory	0.5,10,15,20,25	6490	False	-1.54	0.05	<0.001
Copper	sub-inhibitory	0.5,10,15,20,25	6490	True	-1.35	0.03	<0.001
Lead	sub-inhibitory	0.5,10	2929	False	2.03	0.16	<0.001
Lead	sub-inhibitory	0.5,10	2929	True	2.40	0.10	<0.001
Lead	inhibitory	0.15,20,25,30	2655	False	-1.59	0.17	<0.001
Lead	inhibitory	0.15,20,25,30	2655	True	-0.55	0.15	<0.001

Table 10. Regime-specific slopes (a^* change per 10% of the metal's maximum dose; random effects for strain, run and plate).

Reading.

- Cr: sub-inhibitory doses (0-0.4) raise a^* by +0.9 per 10% of the range. Inhibitory doses lower it (-0.8 at fixed size, -1.1 without size).
- Pb: sub-inhibitory doses (0-10) raise a^* by +2.0 (without size) to +2.4 (at fixed size) per 10%. Inhibitory doses (15-30) lower it (-0.5 at fixed size, -1.6 without size). The inhibitory estimate rests on few wells after the area cutoff (section 10).
- Cu: only the top dose is inhibitory (area ratio 0.49), yet a^* falls steadily (-1.4 per 10% at fixed size). Cu lowers a^* without much reduction in colony size.

7. Size-matched comparison (item 4)

Colonies are placed in five size bins (quantiles of $\ln$ area within each metal, among colonies of at least 2,000 px). The dose effect is estimated inside each bin.

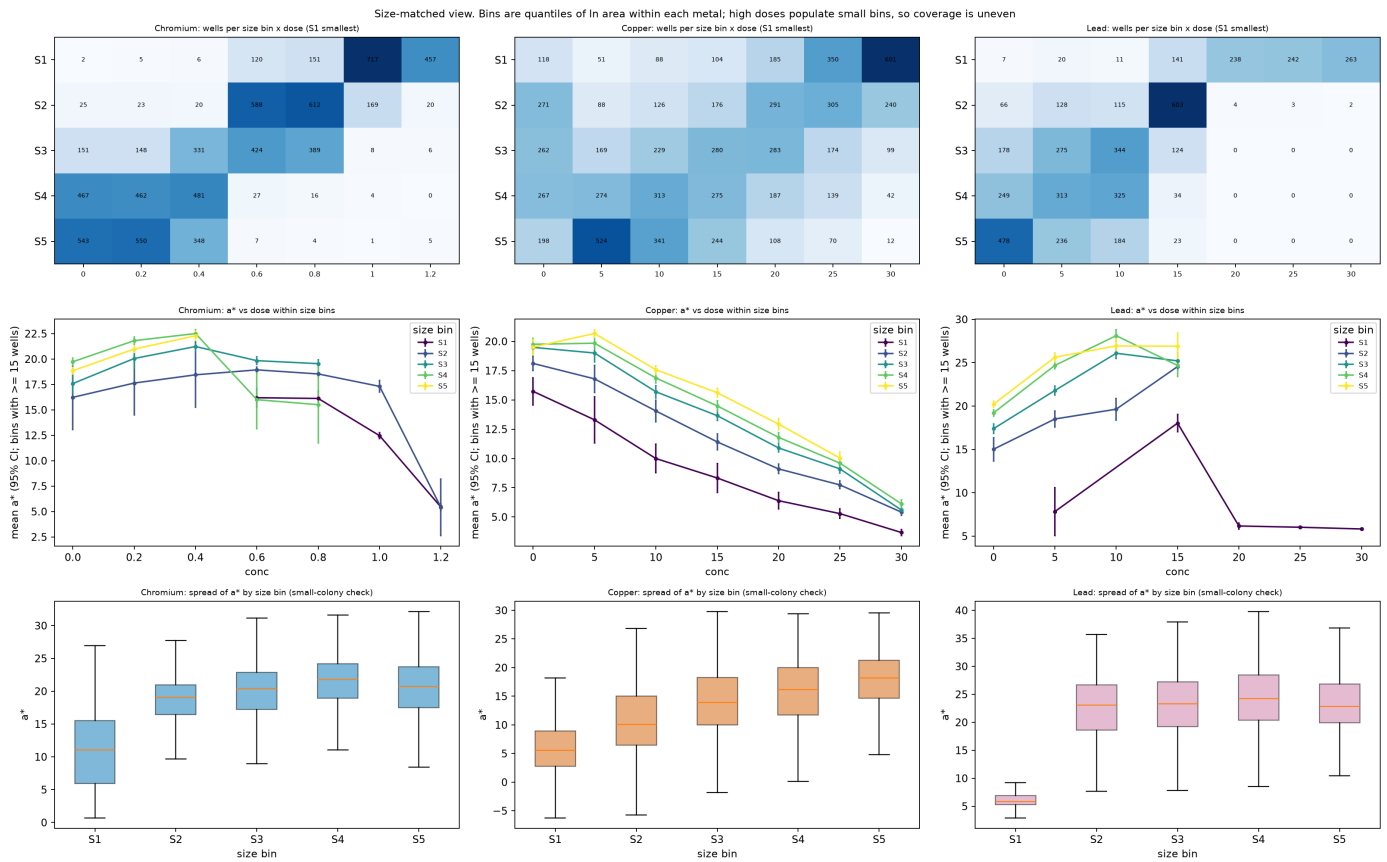


Figure 13. Wells per size bin and dose (top), a^* against dose within bins (middle), and a^* spread by size bin (bottom).

Metal	size_bin	n_wells	n_doses	mean_asd_a	slope_full_range	p	max_dose_s_in	change_over_observed_range
Chromium	S1	1458	7	10.93	5.88	-14.12	1.98	<0.001 1.00 -14.12
Chromium	S2	1457	7	18.32	4.37	-7.25	1.61	<0.001 1.00 -7.25
Chromium	S3	1457	7	19.73	5.20	-4.73	1.15	<0.001 1.00 -4.73
Chromium	S4	1457	6	21.15	4.66	-2.04	1.45	0.16 0.83 -1.70
Chromium	S5	1458	7	20.37	4.72	-10.84	1.99	<0.001 1.00 -10.84
Copper	S1	1497	7	6.36	5.88	-12.41	0.37	<0.001 1.00 -12.41
Copper	S2	1497	7	11.01	6.07	-13.59	0.34	<0.001 1.00 -13.59
Copper	S3	1496	7	14.01	5.68	-14.19	0.35	<0.001 1.00 -14.19
Copper	S4	1497	7	15.88	5.38	-14.07	0.45	<0.001 1.00 -14.07
Copper	S5	1497	7	17.82	4.80	-13.69	0.59	<0.001 1.00 -13.69
Lead	S1	922	7	7.93	5.40	-4.96	1.85	0.01 1.00 -4.96
Lead	S2	921	7	22.25	6.29	8.73	1.81	<0.001 1.00 8.73
Lead	S3	921	4	23.00	5.98	19.47	1.12	<0.001 0.50 9.74
Lead	S4	921	4	24.41	5.91	22.81	1.45	<0.001 0.50 11.40
Lead	S5	921	4	23.11	5.61	22.43	1.51	<0.001 0.50 11.22

Table 11. Dose effect within size bins. $slope_full_range$ is the a^* change over the full dose range; $change_over_observed_range$ multiplies it by the highest dose in the bin (random effects: strain intercept and slope, run, plate).

Reading.

- Cu: a^* falls with dose in every size bin (-12.4 to -14.2 over the full range), and all seven doses are present in every bin. This is the cleanest size-matched result: Cu lowers a^* at the same size.
- Cr: negative in every bin (-14.1 to -2.0). The effect in bin S4 (-2.0) is not significant ($p = 0.16$), and S4 lacks dose 1.2.
- Pb: size and dose stay confounded. The three largest bins contain only doses 0-15. Their slopes (+19 to +23 per full dose range) extrapolate twice beyond the observed range. The change over the observed range is +9.7, +11.4

and +11.2 (last column of Table 11). The smallest bin is negative (-5.0) and the second smallest positive (+8.7). There is no single size-matched Pb estimate.

- **Coverage is uneven.** High doses populate the small bins, and some size-by-dose cells in Cr and Pb hold few wells. Bins are quantiles of pooled ln area, so dose and size remain confounded inside a bin.

8. Selection on baseline, split-half (item 5)

Strains need at least 2 replicate wells at dose 0 and at the top dose. Replicate wells are split at random into halves A and B (1,000 splits). **Naive:** select the top and bottom baseline thirds from all wells and measure the change in the same wells. **Split-half:** select on half A, measure the change in half B.

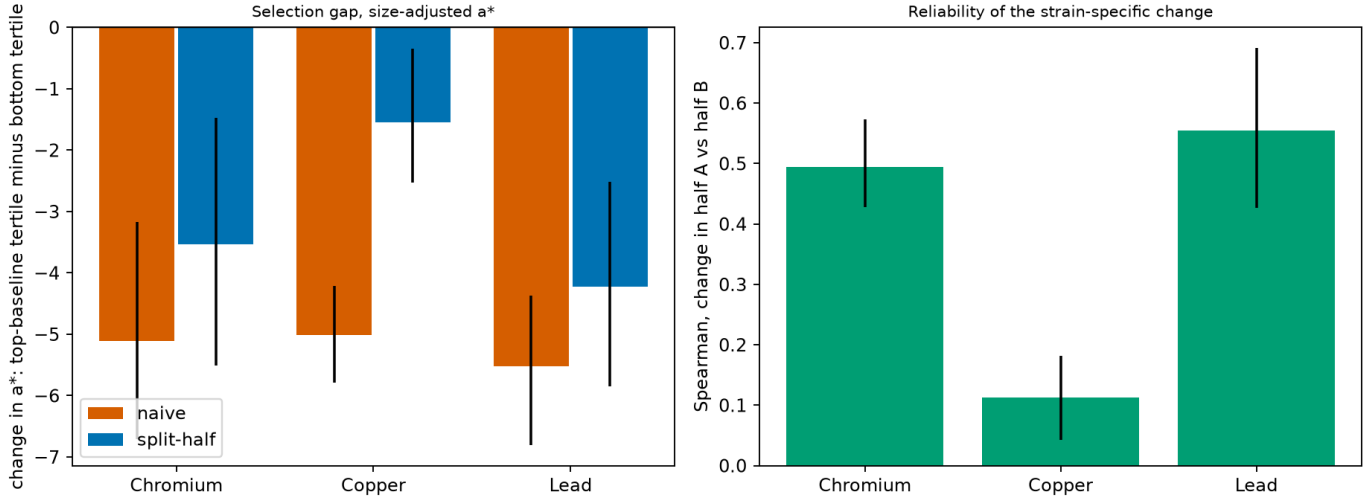


Figure 14. Gap in the change in a^* between the high- and low-baseline thirds (left) and reliability of the strain-specific change (right).

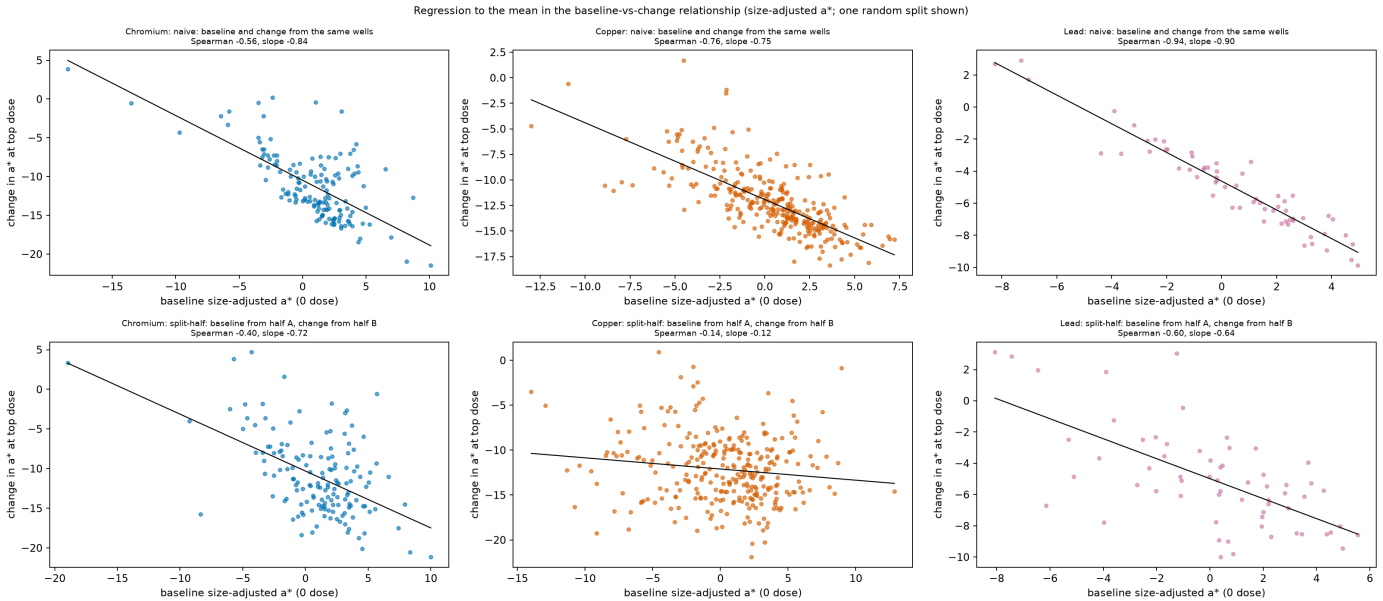


Figure 15. Baseline against change in a^* : same wells (top) and independent halves (bottom).

Metal	top_dose	naive_gap	naive_ci_low	naive_ci_high	split_gap	split_ci_low	split_ci_high	reliability	reliability_B
Chromium	148	-5.11	-6.71	-3.17	-3.53	-5.50	-1.47	0.49	0.60
Copper	298	-5.01	-5.79	-4.21	-1.55	-2.53	-0.35	0.11	0.25
Lead	62	-5.52	-6.80	-4.37	-4.22	-5.85	-2.52	0.55	0.70

Table 12. Size-adjusted a^* . **naive_gap** and **split_gap_mean** are the change in the top-baseline third minus the bottom third (95% CI from a bootstrap over strains). **reliability_change_A_vs_B** is the Spearman correlation of the strain-specific change between the two halves.

Reading.

- Part of the strong negative baseline-vs-change relationship is regression to the mean. The gap shrinks from -5.0 to -1.5 in Cu, from -5.1 to -3.5 in Cr and from -5.5 to -4.2 in Pb.
- In Cu the strain-specific change is barely repeatable (0.11).
- Only 62 Pb strains have enough wells at dose 30 after the area cutoff (Cr 148, Cu 298), so the Pb estimate is uncertain.
- The reliabilities are for halves of 1-2 wells and understate full-data reliability.
- “Size-adjusted” here is the residual of a^* on $\ln$ area from a line fitted to the dose-0 wells of each metal. That differs from the $\ln(\text{area})$ term in the mixed models, and it extrapolates a linear slope to small top-dose colonies.

9. Batch (item 6)

Each metal is analysed separately, with run and plate as random effects (all models above). Each run is a different set of 67-84 strains across all doses, so run and strain set are confounded.

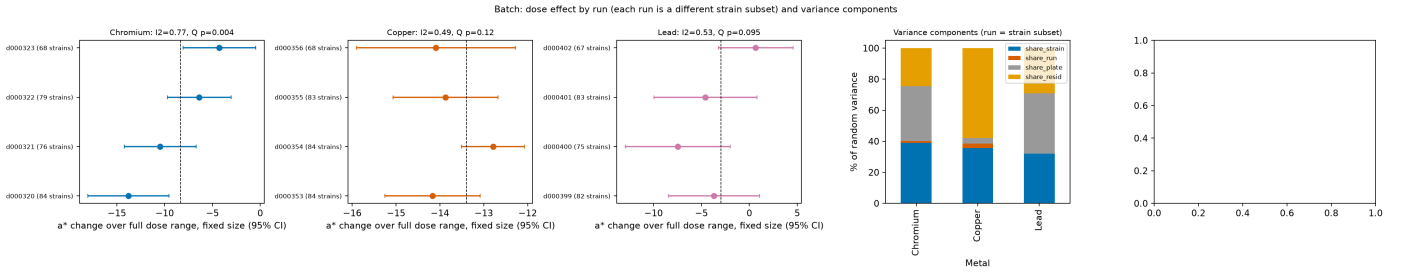


Figure 16. Dose effect at fixed size by run (left three panels) and variance components (right).

Metal	k_runs	pooled_effect	Q	Q_p	I2
Chromium	4	-8.33	13.30	0.00	0.77
Copper	4	-13.40	5.88	0.12	0.49
Lead	4	-2.97	6.37	0.10	0.53

Table 13. Heterogeneity of the dose effect across runs (I2, Cochran Q).

Metal	run	n_wells	n_strains	n_doses	dose_effect	se	p
Chromium	d000320	2018	84	7	-13.78	2.16	<0.001
Chromium	d000321	1879	76	7	-10.47	1.92	<0.001
Chromium	d000322	1877	79	7	-6.38	1.70	<0.001
Chromium	d000323	1513	68	7	-4.27	1.94	0.03
Copper	d000353	2078	84	7	-14.17	0.56	<0.001
Copper	d000354	1957	84	7	-12.79	0.36	<0.001
Copper	d000355	1863	83	7	-13.87	0.61	<0.001
Copper	d000356	1586	68	7	-14.09	0.92	<0.001
Lead	d000399	1412	82	7	-3.70	2.43	0.14
Lead	d000400	1109	75	7	-7.48	2.79	0.01
Lead	d000401	1097	83	7	-4.58	2.75	0.10
Lead	d000402	988	67	7	0.66	1.99	0.74

Table 14. Dose effect (a^* change over the full dose range at fixed size) per run.

Metal	n_runs	share_strain	share_run	share_plate	share_resid	singular
Chromium	4	0.39	0.01	0.35	0.24	False
Copper	4	0.36	0.03	0.04	0.58	False
Lead	4	0.32	0.00	0.39	0.29	True

Table 15. Variance components (shares of random variance).

Reading. Run explains 0-3% of the random variance. Plate explains 35% (Cr) and 39% (Pb), but only 4% in Cu. Only Cr shows clear heterogeneity of the dose effect between runs ($I^2 = 0.77$, $Q p = 0.004$). The Cr effect shrinks from run d000320 to d000323 (-13.8, -10.5, -6.4, -4.3). This could be a run-order effect or a difference between strain subsets. The Cu runs lie within -12.8 to -14.2 and the heterogeneity is not significant ($p = 0.12$). These models have a random strain dose slope.

10. Colony area cutoff (2,000 px)

Very small colonies may have a^* dominated by background pixels. Every image was used to test this. Colonies are small early at every dose. So unstressed small colonies (early) can be compared with stressed small colonies (late) at the same area.

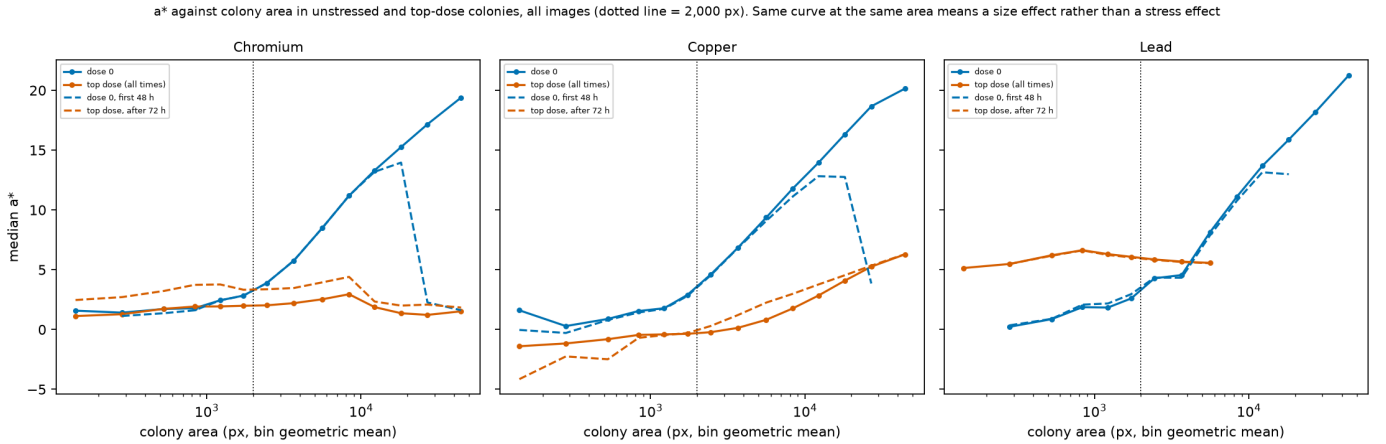


Figure 17. Median a^* against colony area, unstressed (dose 0) and top dose, all images. The dotted line is 2,000 px.

area (px, bin start)	Chromium	Copper	Lead
100	1.6	1.6	
200	1.4	0.3	0.2
400	1.7	0.9	0.9
700	1.8	1.5	1.9
1000	2.4	1.8	1.8
1500	2.8	2.9	2.6
2000	3.9	4.6	4.3
3000	5.8	6.8	4.6
4500	8.5	9.4	8.1
7000	11.2	11.8	11.1
10000	13.3	13.9	13.7
15000	15.3	16.3	15.9
22000	17.2	18.7	18.2
32990	19.4	20.1	21.3

Table 16. Median a^* in area bins for unstressed colonies (dose 0). Bins with fewer than 30 colonies are blank.

Metal	scope	n	change-point (px)	slope_below	slope_above	share of wells below
Chromium	dose 0	20229	1834.85	0.05	5.19	0.04
Chromium	top dose	20245	8000	0.52	-1.67	0.97
Chromium	all doses	147748	2030.97	1.01	5.65	0.20
Copper	dose 0	21281	1744.01	0.91	5.61	0.13
Copper	top dose	19271	4814.78	0.59	2.85	0.42
Copper	all doses	140311	5329.41	1.41	7.17	0.32
Lead	dose 0	15436	3207.49	1.72	6.47	0.27
Lead	top dose	13553	735.66	1.53	-1.06	0.15
Lead	all doses	104024	2897.76	-0.01	7.36	0.55

Table 17. Change-point of a^* against $\ln$ area (piecewise linear fit; area in px where the slope increases).

Reading.

- **a^* depends on area in every metal, even without stress.** At dose 0, median a^* is 0.2-2.4 below 1,500 px, 2.6-2.9 at 1,500-2,000 px, 3.9-4.6 at 2,000-3,000 px, 8-9 at 4,500 px and 19-21 at 33,000 px. The curve is nearly the same in Cr, Cu and Pb. The first-48-h curve (dashed) matches the all-times curve, so area and colony age cannot be separated.
- **Cr and Cu: stressed and unstressed colonies differ little in a^* below about 1,500 px.** At 100-1,500 px the top-dose median is 1.1-1.9 (Cr) and -1.4 to -0.4 (Cu), against 1.4-2.4 and 0.3-1.8 at dose 0. The gap is at most about 2 a^* units there (Cu), against 10 or more above 5,000 px. Above 1,500-2,000 px stressed colonies are clearly lower.
- **Pb: stressed colonies of the same small size have higher a^* than unstressed ones.** Pb top-dose a^* is 5.1-6.6 at 100-1,500 px, against 0.2-1.9 at dose 0. So the Pb floor near 6 is partly a stress effect and not only a pixel effect.
- **The change-point estimates differ by metal** (Table 17: 1,700-1,800 px for Cr and Cu at dose 0; 3,200 px for Pb). The slope below the change-point is 0.05 (Cr) and 0.9 (Cu) but 1.7 for Pb, against 5-6.5 above, so “flat” holds for Cr and Cu and only “low and slowly rising” for Pb. The per-bin medians show a step at about 1,500-2,000 px in all three metals. The 2,000 px value was therefore chosen from the data and applied to all metals. It is assumed, not tested, for Pb.
- **The cutoff does not remove the size confound.** a^* keeps rising with area above 2,000 px, so every model still includes a size term.

What the cutoff removes (share of wells in the unfiltered table that are lost):

Metal	dose	wells, unfiltered	wells, main	retained_ %
Chromium	0	1197	1188	99
Chromium	0.2	1192	1188	100
Chromium	0.4	1188	1186	100
Chromium	0.6	1171	1166	100
Chromium	0.8	1179	1172	99
Chromium	1	1129	899	80
Chromium	1.2	1052	488	46
Copper	0	1138	1116	98
Copper	5	1124	1106	98
Copper	10	1106	1097	99
Copper	15	1107	1079	97
Copper	20	1078	1054	98
Copper	25	1061	1038	98
Copper	30	1038	994	96
Lead	0	1023	978	96
Lead	5	1025	972	95
Lead	10	1013	979	97
Lead	15	1004	925	92
Lead	20	891	242	27
Lead	25	924	245	27

Metal	dose	wells, unfiltered	wells, main	retained_ %
Lead	30	875	265	30

Table 18. Wells in the unfiltered table and in the main table, by metal and dose. Most losses are at the top doses of Cr and Pb.

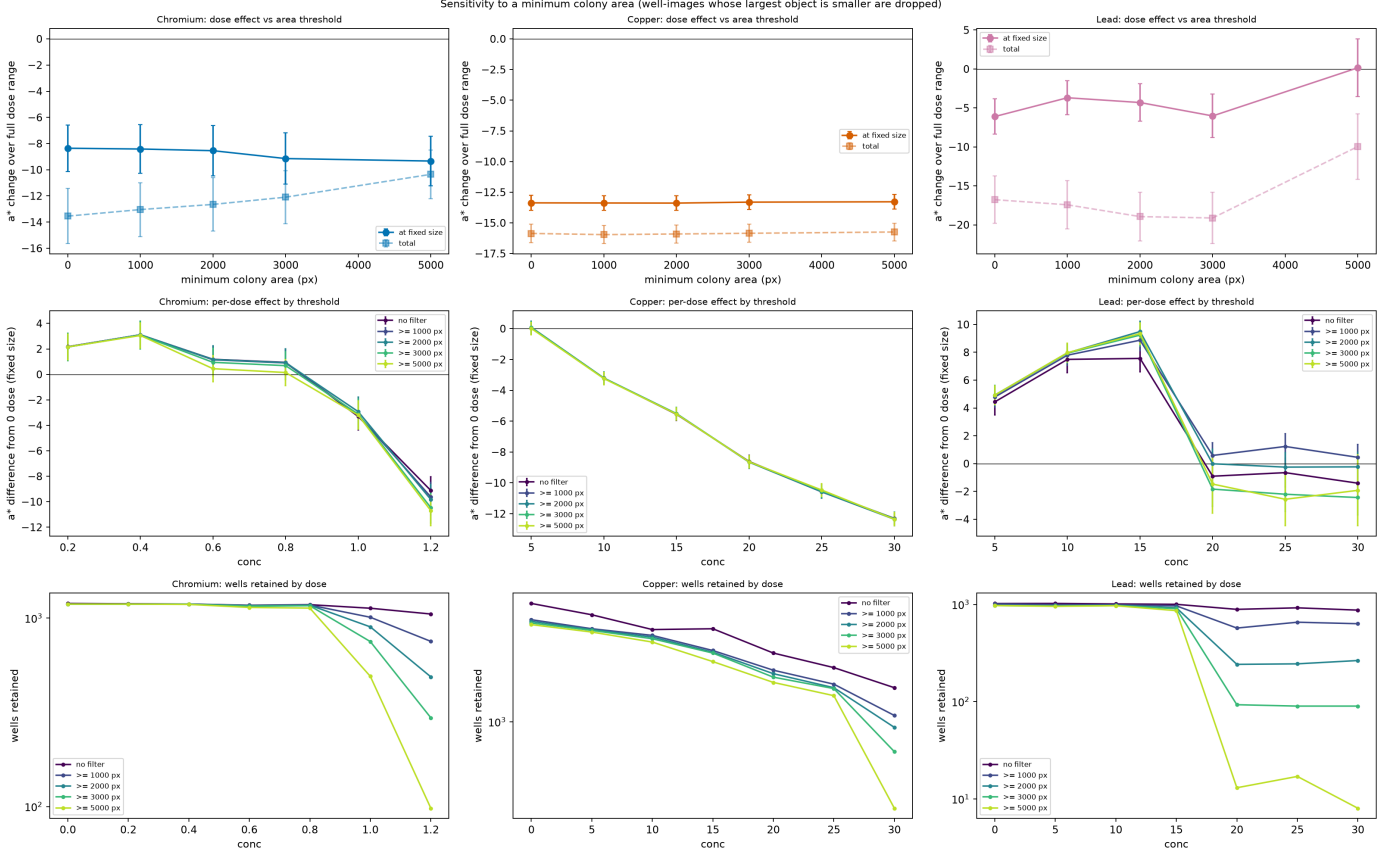


Figure 18. Sensitivity to the area cutoff: dose effect against threshold (top), per-dose effect at fixed size (middle), wells retained by dose (bottom). Threshold 0 is the unfiltered run.

Metal	0	1000	2000	3000	5000
Chromium	-8.36	-8.41	-8.54	-9.14	-9.33
Copper	-13.37	-13.38	-13.39	-13.31	-13.28
Lead	-6.10	-3.69	-4.31	-6.03	0.15

Table 19. Dose effect on a^* (at fixed size) by minimum colony area (0 = no filter).

dose	effect @0	effect @1000	effect @2000	effect @3000	effect @5000	wells @0	wells @1000	wells @2000	wells @3000	wells @5000
0						1197	1189	1188	1188	1187
0.2	2.2	2.2	2.2	2.2	2.2	1192	1190	1188	1187	1187
0.4	3.1	3.1	3.1	3.1	3.1	1188	1187	1186	1185	1183
0.6	1.2	1.1	1.2	0.9	0.4	1171	1168	1166	1158	1140
0.8	0.9	0.9	0.9	0.7	0.1	1179	1174	1172	1164	1131
1	-3.3	-3.2	-2.9	-3.1	-3.2	1129	1011	899	751	491
1.2	-9.1	-9.6	-9.8	-10.5	-10.7	1052	753	488	296	98

Table 20. Chromium: a^* difference from 0 dose at fixed size, by threshold, and wells retained.

dose	effect @0	effect @1000	effect @2000	effect @3000	effect @5000	wells @0	wells @1000	wells @2000	wells @3000	wells @5000
0						1023	1015	978	975	974
5	4.4	4.8	4.9	4.9	4.9	1025	997	972	962	958
10	7.5	7.8	8	7.9	8	1013	994	979	975	971
15	7.6	8.9	9.5	9.2	9.3	1004	968	925	906	864
20	-0.9	0.6	-0	-1.8	-1.5	891	572	242	93	13
25	-0.6	1.2	-0.3	-2.2	-2.6	924	657	245	90	17
30	-1.4	0.4	-0.2	-2.4	-1.9	875	634	265	90	8

Table 21. Lead: a^* difference from 0 dose at fixed size, by threshold, and wells retained.

Metal	min_area	top_dose	n_strains_top	top_dose_mean	atop_dose_sd_across_strains	median_area_top_px
Chromium	0	1.20	311	4.43	2.52	1728.99
Copper	0	30.00	314	4.06	2.64	16541.21
Lead	0	30.00	307	6.48	1.09	1429.57
Chromium	1000	1.20	284	4.95	3.04	2405.72
Copper	1000	30.00	309	4.34	2.38	16974.50
Lead	1000	30.00	291	6.32	1.17	1794.31
Chromium	2000	1.20	241	5.12	3.57	3281.38
Copper	2000	30.00	308	4.45	2.35	17287.13
Lead	2000	30.00	189	5.87	0.77	2712.98
Chromium	3000	1.20	177	5.53	4.11	4101.66
Copper	3000	30.00	307	4.52	2.34	17960.48
Lead	3000	30.00	81	5.67	0.56	3476.00
Chromium	5000	1.20	74	7.13	6.15	6516.56
Copper	5000	30.00	306	4.66	2.37	19181.04
Lead	5000	30.00	8	5.28	0.98	6610.00

Table 22. Strain-mean a^* at the top dose by threshold.

Reading.

- **Cu is unaffected** (-13.4 to -13.3).
- **Survivor selection.** After the cutoff the top-dose estimates describe larger colonies. In Cr, 241 strains remain at the top dose (311 without a cutoff) and the median top-dose area rises from 1,729 to 3,281 px (Table 22). In Pb, 27-30% of the unfiltered wells at doses 20-30 remain. Compare the main and unfiltered numbers with this in mind.
- **Cr: the top-dose drop is not a small-colony artifact.** The effect at dose 1.2 is -9.1 with no filter and -10.7 with colonies of at least 5,000 px. The overall Cr dose effect stays between -8.4 and -9.3.
- **Pb: the rise at doses 5-15 holds or strengthens** (+4.9, +8.0 and +9.5 at 2,000 px, against +4.4, +7.5 and +7.6 with no filter). At doses 20-30 only 242-265 wells remain at 2,000 px and 8-17 at 5,000 px, so the Pb inhibitory range cannot be tested. The overall Pb dose effect is unstable across cutoffs and should not be quoted as one number.
- **A floor remains in Pb.** Among Pb colonies of at least 3,000 px at dose 30 (81 strains), strain-mean a^* is 5.7 with SD 0.56 across strains.

11. Stratified by species and ploidy group

Pure haploid *R. mucilaginosa* is shown alone. Each other group with at least 5 strains in at least two metals is shown alone against the reference as a grey line. Lines show strain means with 95% CI across strains.

species	Chromium	Copper	Lead
<i>Rhodotorula mucilaginosa</i>	173	173	172
<i>Rhodotorula</i> aff. <i>mucilaginosa</i>	7	7	7
<i>Rhodotorula dairenensis</i>	8	10	10
<i>Rhodotorula diobovata</i>	9	9	9

species	Chromium	Copper	Lead
Rhodotorula frigidialcoholis	5	5	5
Rhodotorula mucilaginosa hybrid diploid	33	33	33
Rhodotorula paludigena	12	18	9
Rhodotorula sp_clade_I	8	9	9
Rhodotorula sphaerocarpa	6	7	7
Rhodotorula taiwanensis	6	6	6
Rhodotorula toruloides	10	10	10

Table 23. Strains per group and metal.

11a. Pure haploid *R. mucilaginosa* alone

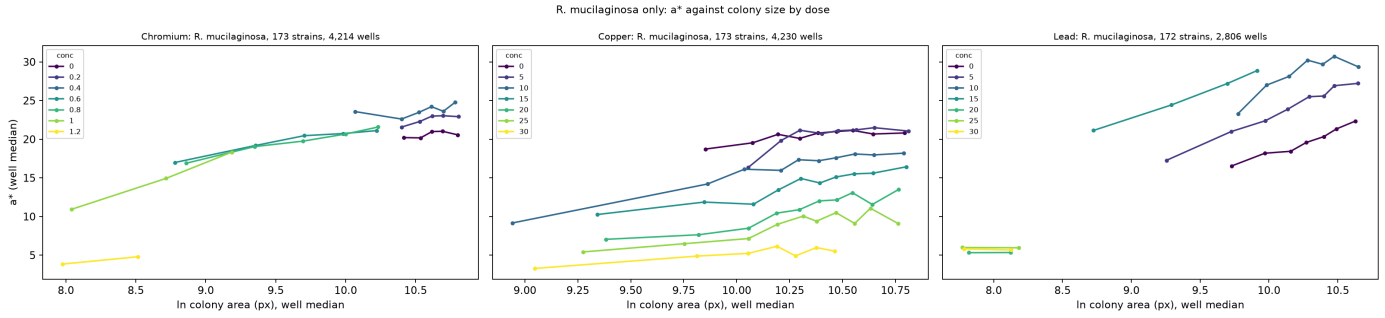


Figure 19. Pure haploid *R. mucilaginosa* only: a^* against colony size by dose.

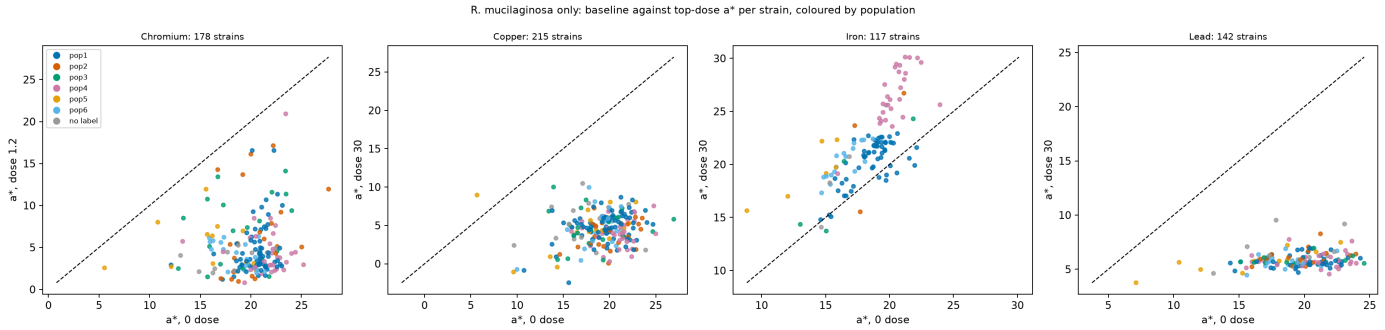


Figure 20. Pure haploid *R. mucilaginosa* only: baseline against top-dose a^* per strain.

11b. Each other group alone

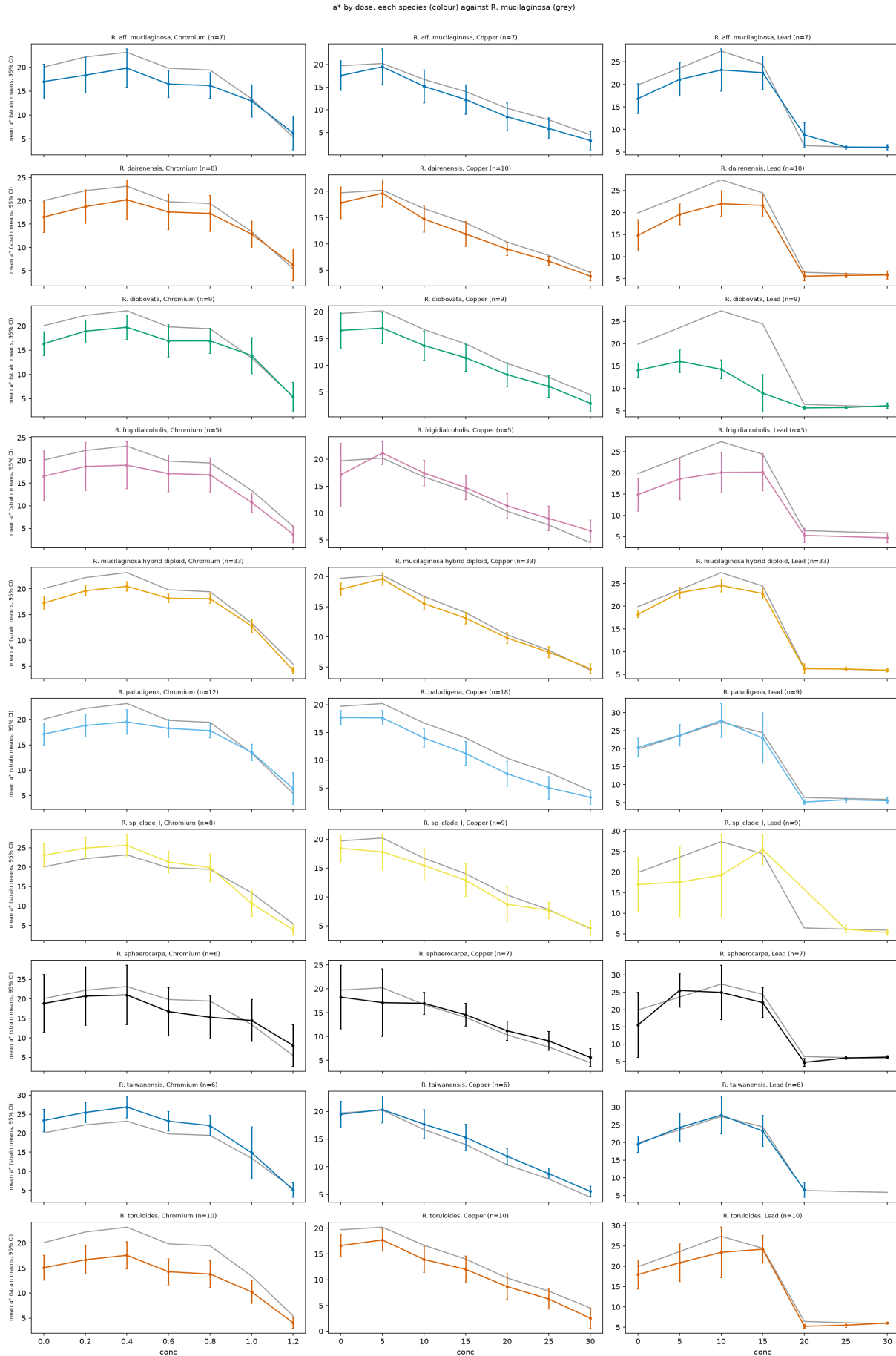


Figure 21. a^* by dose for each group (colour) against pure haploid *R. mucilaginosa* (grey).

Colony size (ln area) by dose, each species (colour) against *R. mucilaginosa* (grey)

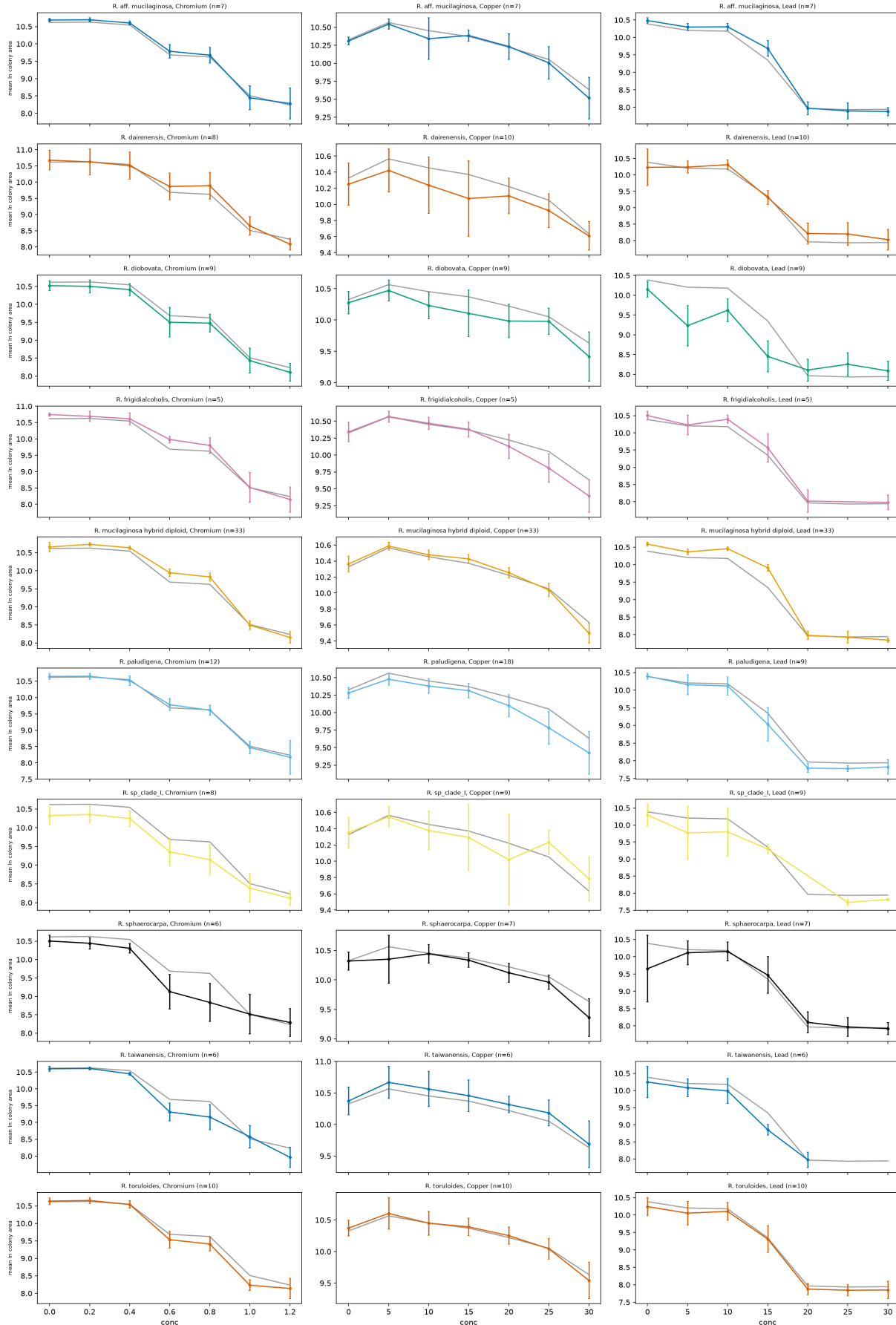


Figure 22. Colony size (ln area) by dose for each group against pure haploid R. mucilaginosa.

Mixed-model dose effect on a^* at fixed size, each species (colour) against *R. mucilaginosa* (grey): 95% CI



Figure 23. Dose effect on a^* at fixed size for each group against pure haploid *R. mucilaginosa* (95% CI). One model per metal with a dose-by-group interaction and one common size slope, as in Table 5.

stratum	Chromium	Copper	Lead
Other species (≥ 5 strains, pooled)	-11.1	-15.4	-16.4
<i>R. aff. mucilaginosa</i>	-7.2	-16.5	-12.2
<i>R. dairenensis</i>	-9.0	-16.7	-10.0
<i>R. diobovata</i>	-9.8	-14.9	-11.2
<i>R. frigidialcoholis</i>	-9.1	-12.5	-8.9
<i>R. mucilaginosa</i>	-13.4	-16.5	-20.8
<i>R. mucilaginosa</i> hybrid diploid	-10.8	-15.0	-17.6
<i>R. paludigena</i>	-8.6	-16.9	-19.7
<i>R. sp_clade_I</i>	-16.4	-15.6	-17.4
<i>R. sphaerocarpa</i>	-8.6	-16.4	-13.4
<i>R. taiwanensis</i>	-10.7	-14.9	-13.3
<i>R. toruloides</i>	-9.4	-15.5	-16.4

Table 24. Total dose effect on a^* over the full dose range (no size term), by group. Random intercepts for strain and plate only (plate is nested in run), so the intervals are optimistic. “Other species” pools all groups with at least 5 strains except pure haploid *R. mucilaginosa*. A per-group size slope cannot be estimated reliably in small strata, because dose and size are confounded, so fixed-size effects by group come from the shared-slope models (Table 5, Figure 23).

Baseline against top-dose a^* per strain, each species (colour) over *R. muclaginoso* (grey): dashed = no change

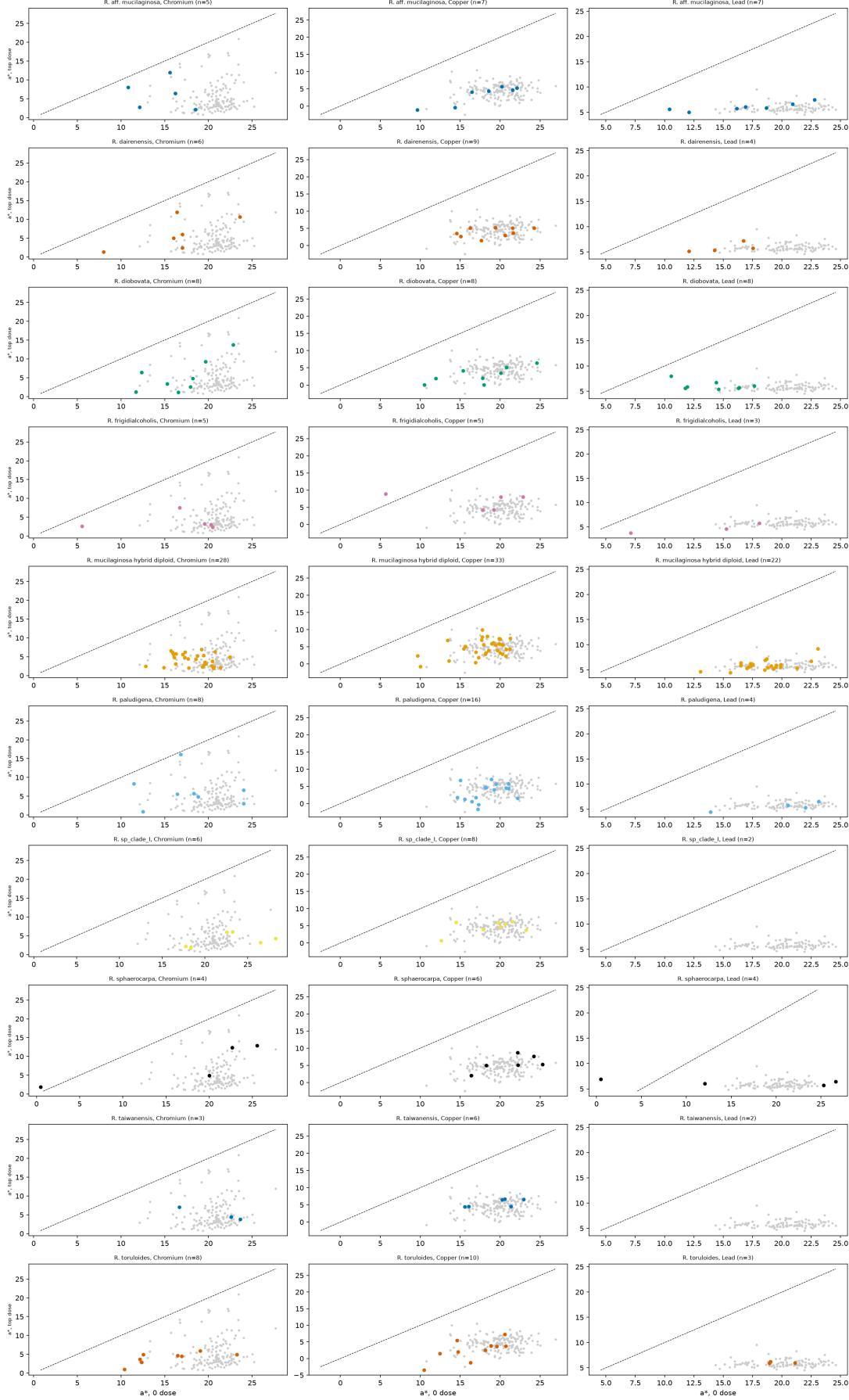


Figure 24. Baseline against top-dose a^* per strain, each group (colour) over pure haploid *R. mucilaginosa* (grey).

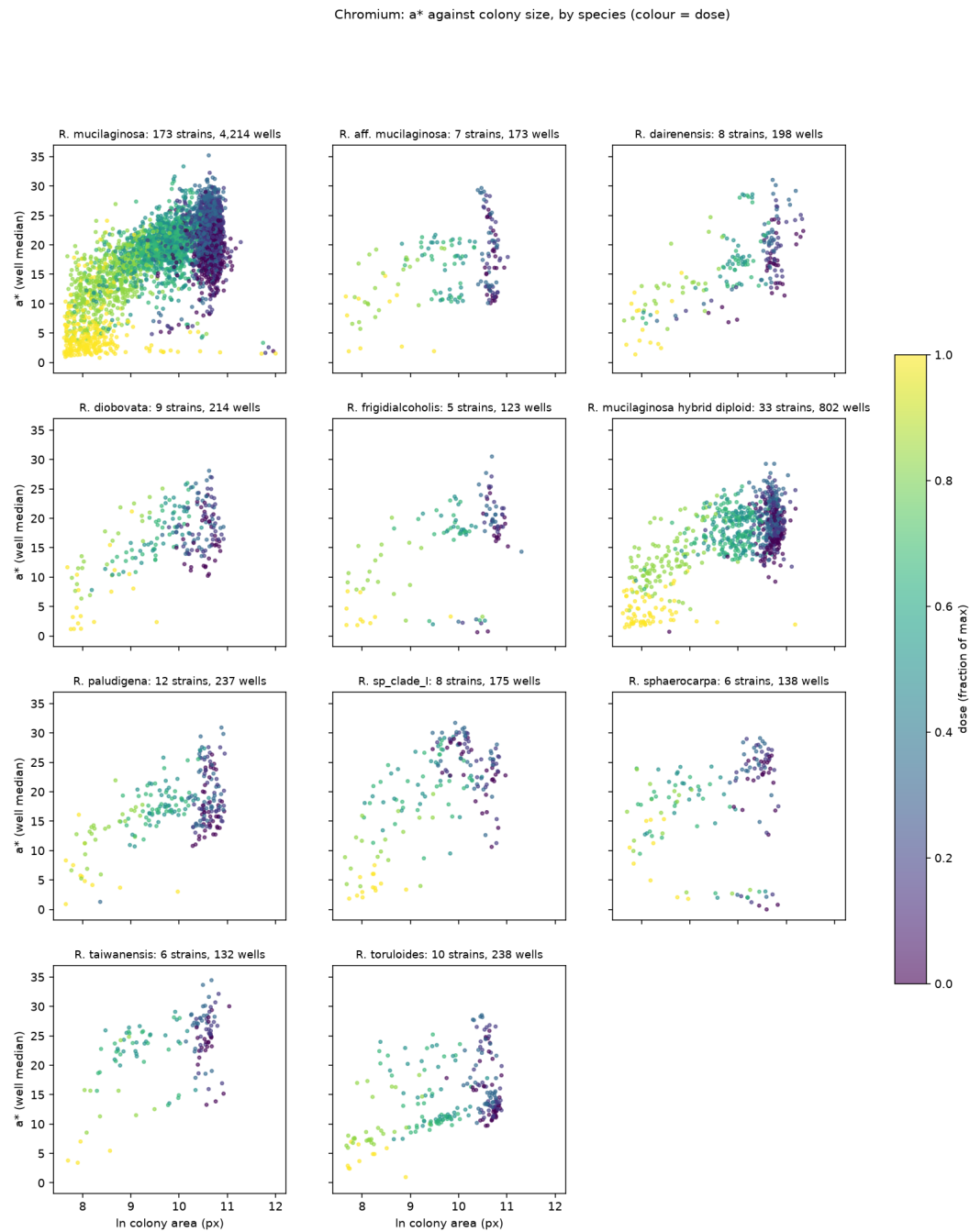


Figure 25. Chromium: a^* against colony size by group (colour = dose).

Copper: a^* against colony size, by species (colour = dose)

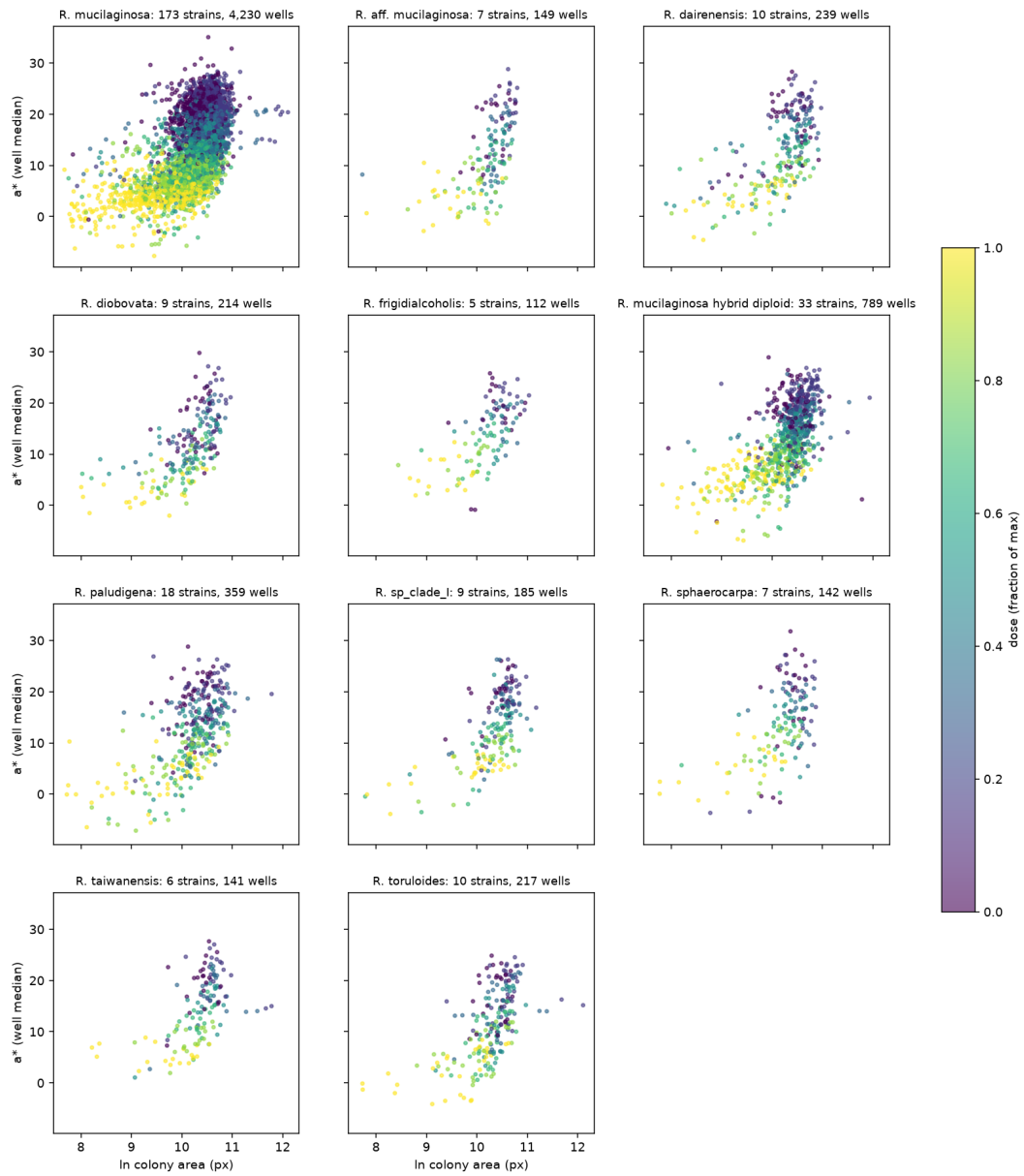


Figure 26. Copper: a^* against colony size by group (colour = dose).

Lead: a^* against colony size, by species (colour = dose)

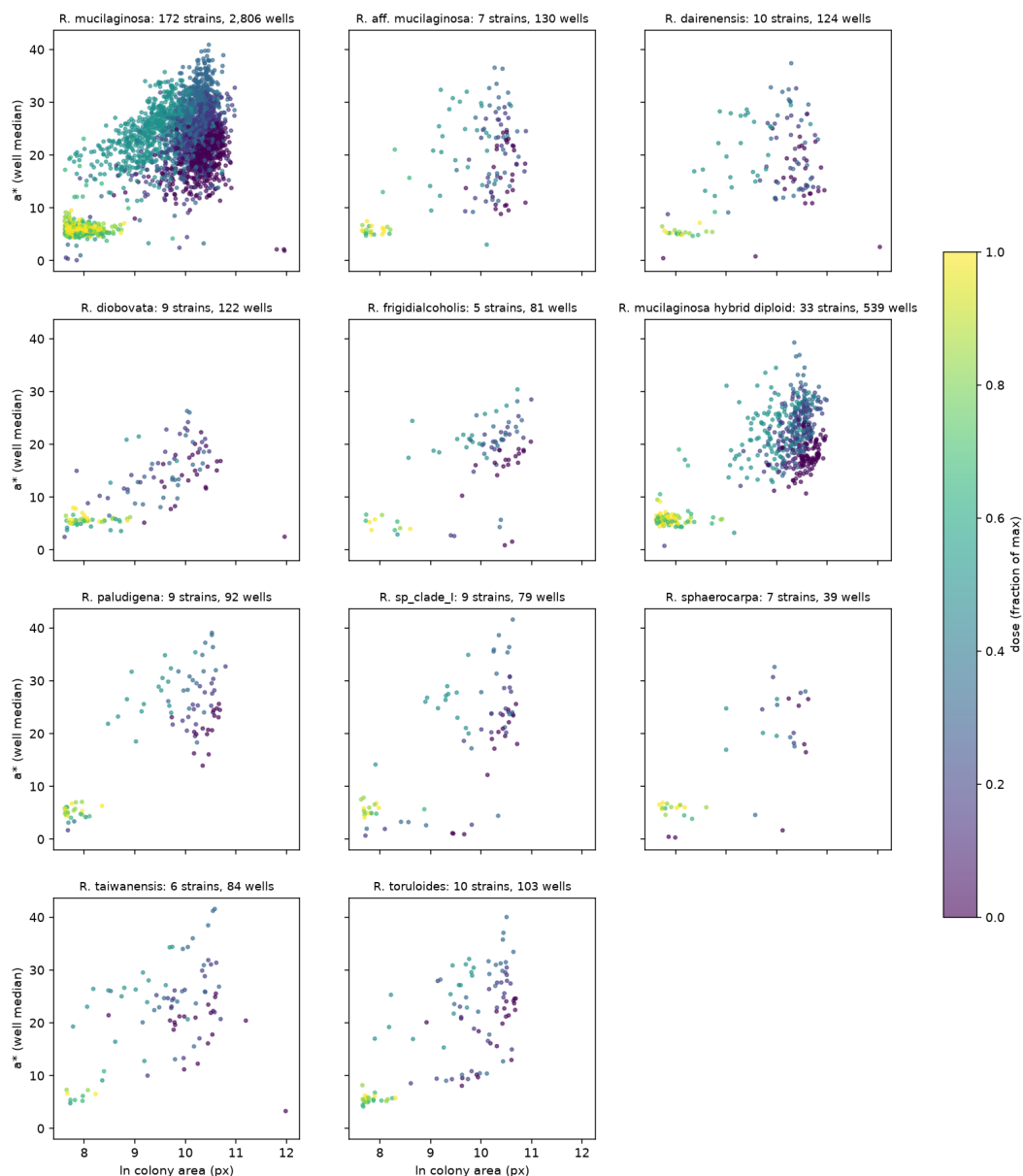


Figure 27. Lead: a^* against colony size by group (colour = dose).

Reading.

- **Cu is similar in most groups.** The fixed-size slope over the full range is -12.0 to -14.2 in 9 of the 10 other groups (Table 5), against -14.2 for the reference. *R. frigidialcoholis* is weaker (-9.7). Hybrids are -12.2.
- **Total effects** (no size term, Table 24) range from -7.2 to -16.4 in Cr, -12.5 to -16.9 in Cu and -8.9 to -20.8 in Pb. Pure haploid *R. mucilaginosa* has the largest Pb total effect (-20.8).
- **Hybrid diploids and aff. mucilaginosa respond less than the reference in Cr** (-10.8 and -7.2 against -13.4 in Table 24) **and in Pb** (-17.6 and -12.2 against -20.8).
- **Single-group curves are noisy.** Most groups have 5-18 strains, so the confidence intervals are wide.

12. b^* and morphology

Colour traits: L^* , a^* , b^* , chroma (square root of $a^{*2} + b^{*2}$) and hue angle ($\text{atan2}(b^*, a^*)$ in degrees; 0 is red and 90 is yellow). Morphology: circularity, solidity, eccentricity, compactness, extent and aspect ratio (major / minor axis). Wells and window are the main dataset. For the models, each trait is scaled by the mean and SD of the 0-dose wells of its metal, so effects are in SD units of the unstressed wells. Lines show strain means with 95% CI across strains, for pure haploid *R. mucilaginosa* and all other groups pooled.

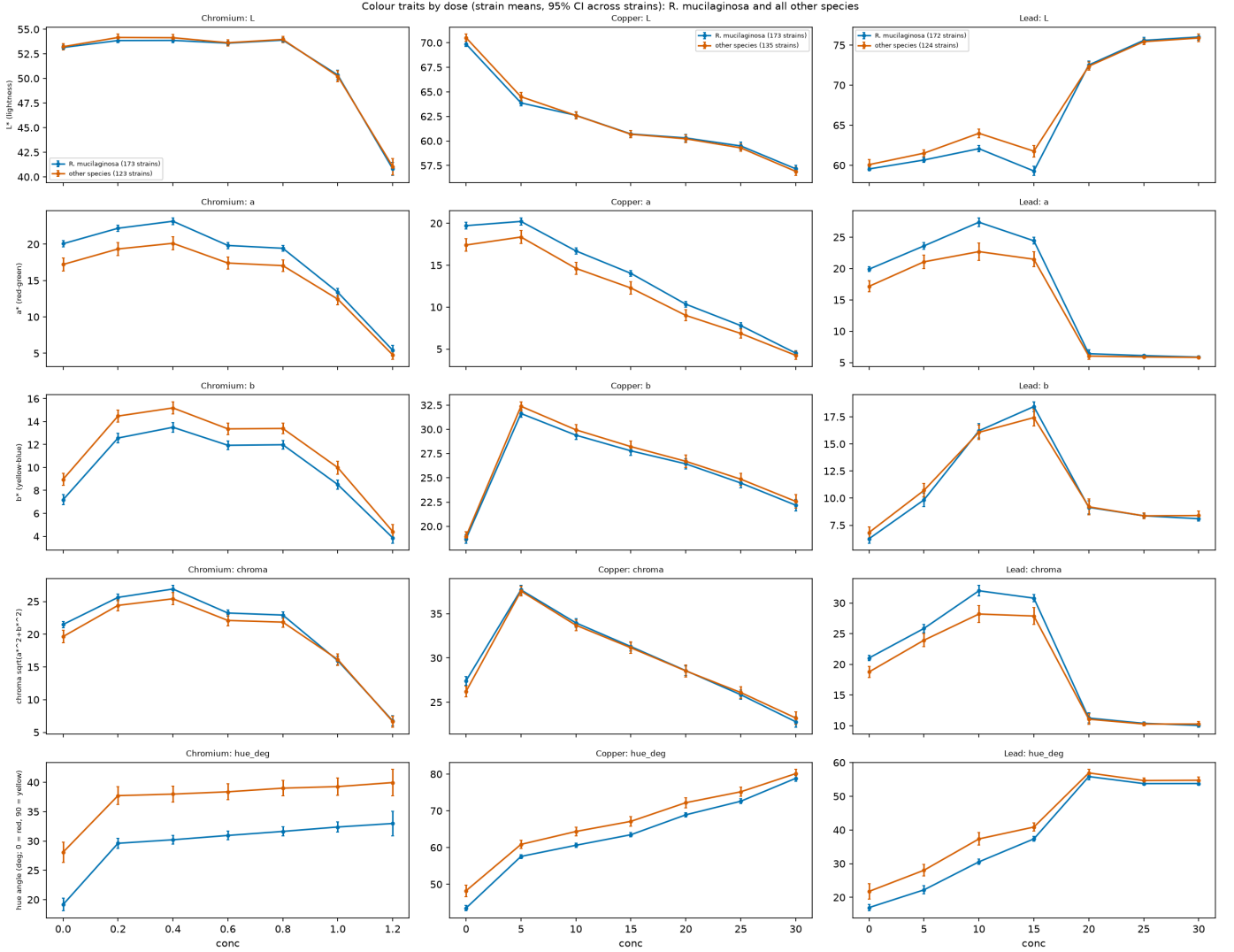


Figure 28. Colour traits by dose: pure haploid *R. mucilaginosa* and other groups.

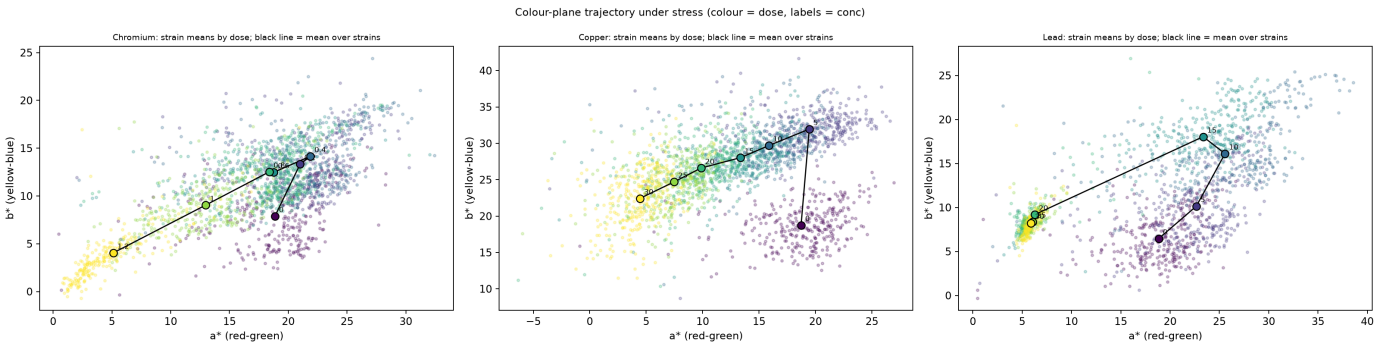


Figure 29. Trajectory in the a^*-b^* plane under stress (colour = dose; labels give the concentration).

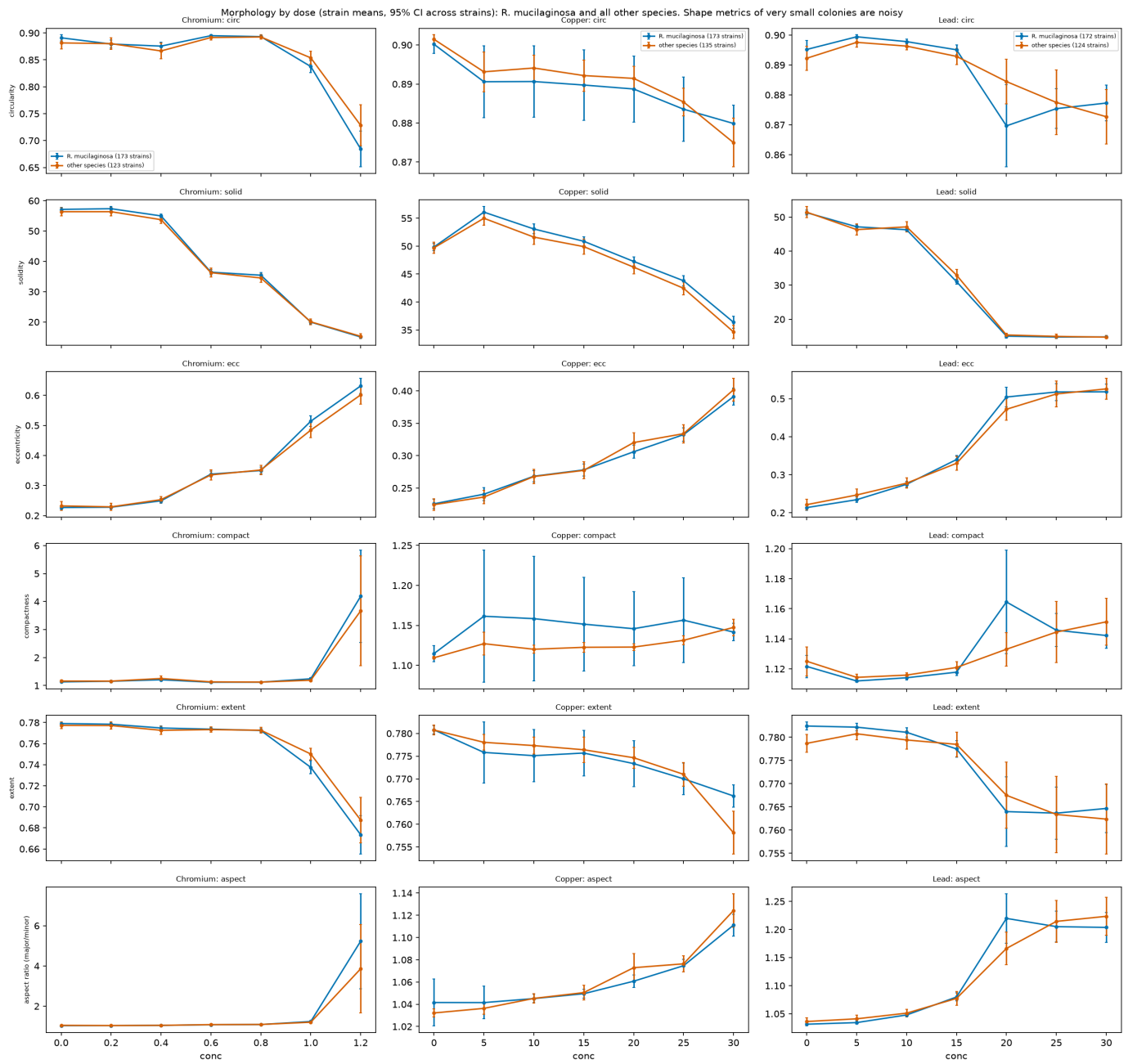


Figure 30. Morphology by dose: pure haploid *R. mucilaginosa* and other groups. Shape metrics of very small colonies are noisy.

Mixed-model dose effect (0 to max dose) on colour and morphology traits, in SD units. Stars: $p < .05$, $.01$, $.001$ (unadjusted)

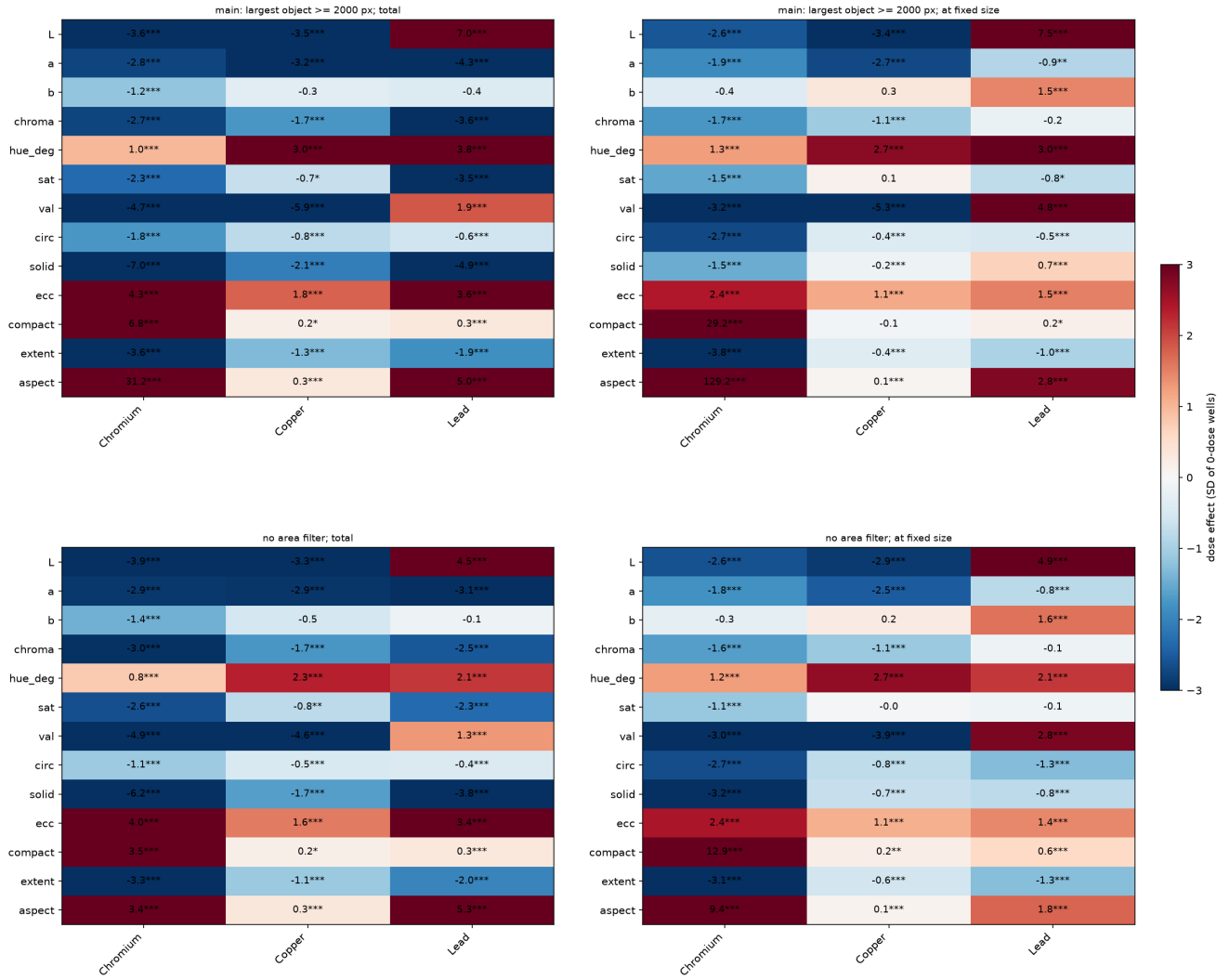


Figure 31. Mixed-model dose effect on colour and morphology traits in SD units (stars: unadjusted $p < 0.05$, 0.01 , 0.001). Rows of panels: main dataset and no area filter. Columns: total effect and effect at fixed size.

trait	Chromium	Copper	Lead
L	-3.6	-3.5	7.0
a	-2.8	-3.2	-4.3
b	-1.2	-0.3	-0.4
chroma	-2.7	-1.7	-3.6
hue_deg	1.0	3.0	3.8
sat	-2.3	-0.7	-3.5
val	-4.7	-5.9	1.9
circ	-1.8	-0.8	-0.6
solid	-7.0	-2.1	-4.9
ecc	4.3	1.8	3.6
compact	6.8	0.2	0.3
extent	-3.6	-1.3	-1.9
aspect	31.2	0.3	5.0

Table 25. Dose effect over the full dose range, main dataset, total effect (SD of the unstressed wells).

trait	Chromium	Copper	Lead
L	-2.6	-3.4	7.5
a	-1.9	-2.7	-0.9
b	-0.4	0.3	1.5
chroma	-1.7	-1.1	-0.2
hue_deg	1.3	2.7	3.0
sat	-1.5	0.1	-0.8
val	-3.2	-5.3	4.8
circ	-2.7	-0.4	-0.5
solid	-1.5	-0.2	0.7
ecc	2.4	1.1	1.5
compact	29.2	-0.1	0.2
extent	-3.8	-0.4	-1.0
aspect	129.2	0.1	2.8

Table 26. Same, at fixed colony size.

dose	a	b	lnA
0.0	18.7	18.8	10.3
5.0	19.4	32.0	10.5
10.0	15.9	29.7	10.4
15.0	13.3	28.0	10.4
20.0	9.9	26.6	10.2
25.0	7.5	24.7	10.0
30.0	4.4	22.4	9.6

Table 27. Copper: strain-mean a^* , b^* and $\ln$ area by dose (main dataset).

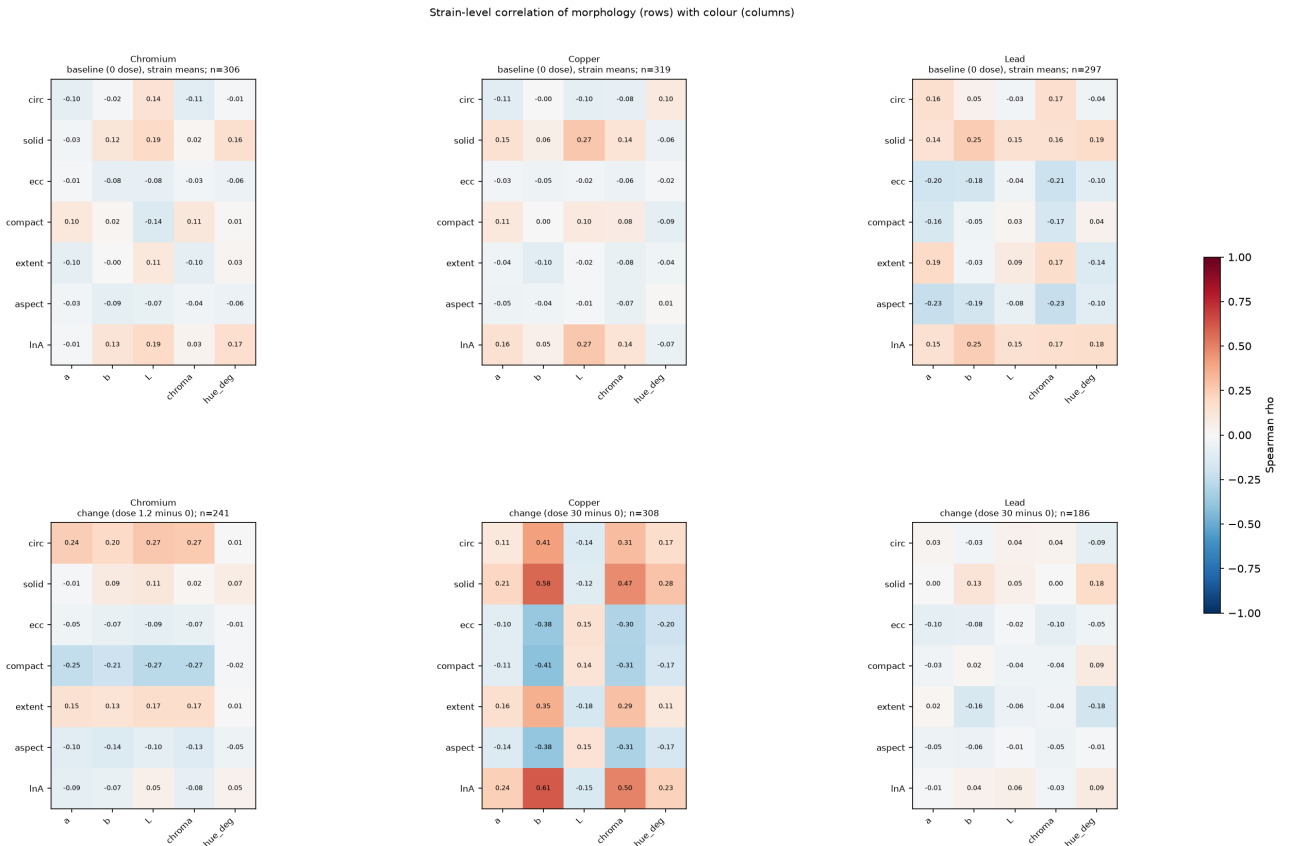


Figure 32. Strain-level Spearman correlation of morphology (rows) with colour (columns): baseline (top) and change at the top dose (bottom).

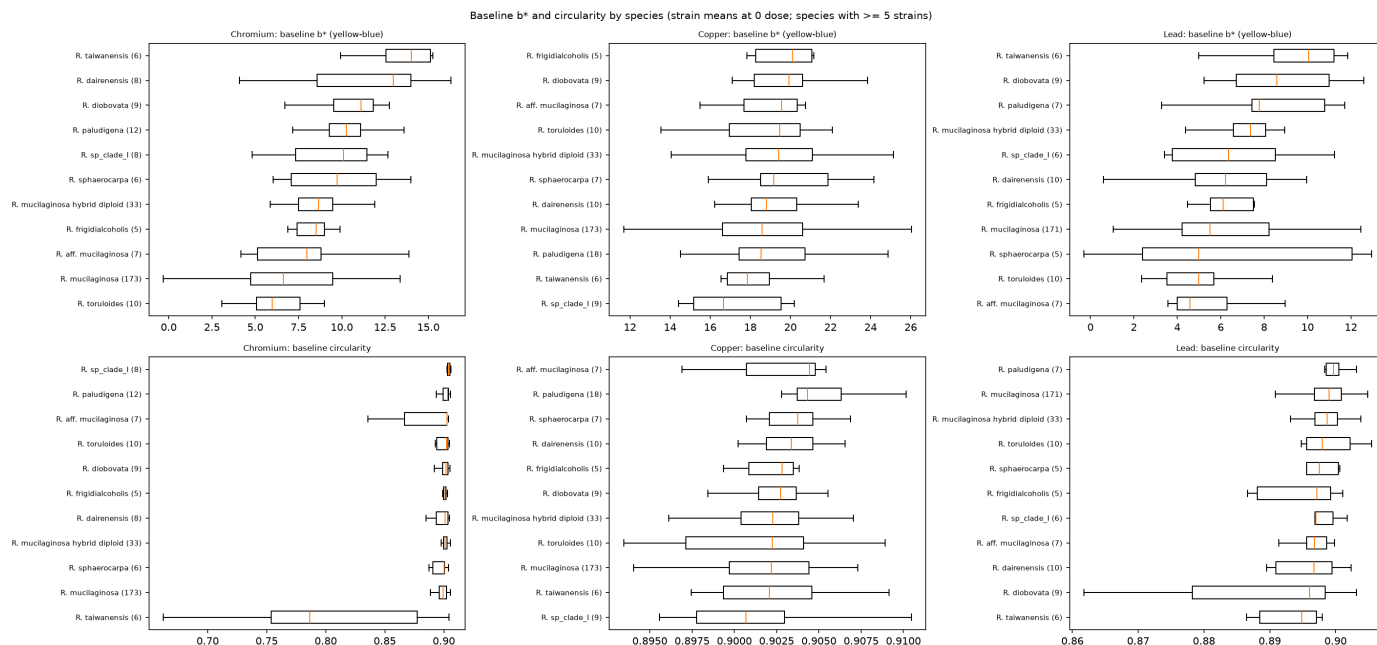


Figure 33. Baseline b^* and circularity by group (strain means at 0 dose).

Reading.

- **b^* moves much less than a^*** (total effect: Cr -1.2, Cu -0.3, Pb -0.4 SD; a^* is -2.8, -3.2, -4.3).
- **The hue angle rises in all three metals** (+1.0 Cr, +3.0 Cu, +3.8 Pb SD, total). Colour shifts from red toward yellow with stress, mainly because a^* falls faster than b^* .
- **Lightness (L^*) falls in Cr and Cu and rises in Pb** (-3.6, -3.5 and +7.0 SD, total). Pb colonies at doses of 20 and above are tiny and pale.
- **Colonies become less compact under stress.** Solidity falls (Cr -7.0 SD, Pb -4.9, Cu -2.1) and eccentricity rises (Cr +4.3, Pb +3.6, Cu +1.8). The effect is largest in the two metals that inhibit growth most. At fixed size most of the solidity effect disappears (Cr -1.5, Cu -0.2, Pb +0.7), so it follows colony size.
- **Do not read the Cr aspect-ratio and compactness effects as magnitudes** (+31 and +129 SD for aspect ratio, total and fixed size; +29 SD for compactness at fixed size). The unstressed wells have almost no variance in those traits, so the SD scale blows up. Only the direction (more elongated, less compact) is meaningful.
- **Cu b^* jumps from 18.8 at dose 0 to 32.0 at dose 5** (Table 27) while a^* (18.7 to 19.5) and $\ln$ area (10.3 to 10.5) barely change. This looks like a plate or media effect between the dose-0 plate and the treated plates, not biology. Cu b^* and hue results should be read with that in mind.
- Morphology of very small colonies is unreliable (few pixels). The main dataset already drops colonies below 2,000 px.

13. Not analysed: Iron and Zinc

Iron and Zinc are held out (D-56). The ingested Parquet files hold part of the plates only (Iron 62 of 120 plates; Zinc 9 of 119). The earlier Fe and Zn results, including the Zinc rescue, are withdrawn from this report. The full-plate tables are needed before these metals are analysed. See [analysis/heavy_metal_data_problems/HEAVY_METAL_DATA_PROBLEMS.md](#).

14. Sensitivity: results without the area cutoff

The original analysis used no area cutoff. Key numbers from that run (tables in [report/sensitivity_unfiltered/](#)):

Metal	model	dose_effect unfiltered	dose_effect main
Chromium	total	-13.54	-12.65
Chromium	at fixed size	-8.36	-8.54
Copper	total	-15.86	-15.91

Metal	model	dose_effect unfiltered	dose_effect main
Copper	at fixed size	-13.37	-13.39
Lead	total	-16.77	-18.95
Lead	at fixed size	-6.10	-4.31

Table 28. Dose effect (0 to top dose) in the unfiltered run and in the main run.

quantity	unfiltered	main
Chromium: omnibus F, baseline size adjusted	8.16	8.05
Copper: omnibus F, baseline size adjusted	2.96	2.98
Lead: omnibus F, baseline size adjusted	8.72	8.07
Chromium: split-half gap (size-adjusted)	-5.48	-3.53
Chromium: reliability of strain change	0.54	0.49
Copper: split-half gap (size-adjusted)	-1.68	-1.55
Copper: reliability of strain change	0.13	0.11
Lead: split-half gap (size-adjusted)	-4.20	-4.22
Lead: reliability of strain change	0.31	0.55
Chromium: omnibus F, species by dose	10.98	9.94
Chromium: strains in the split-half analysis	289.00	148.00
Copper: omnibus F, species by dose	3.39	3.14
Copper: strains in the split-half analysis	303.00	298.00
Lead: omnibus F, species by dose	13.80	4.75
Lead: strains in the split-half analysis	273.00	62.00

Table 29. Other key numbers, unfiltered and main.

Reading.

- **Cu and Cr are about the same in both runs** (Table 28).
- **Pb changes most.** The Pb dose-effect estimate moves from -6.1 (unfiltered) to -4.3 (main).
- **The comparison is not strictly like with like.** The split-half samples shrink (Cr 289 to 148 strains, Pb 273 to 62, because few strains keep at least 2 top-dose wells), so those rows also reflect fewer strains. Run heterogeneity is not compared: the unfiltered chain used the same slope models, but the number of wells per run changes with the cutoff.
- Pagel’s lambda is not compared between runs.

Limits

- Dose units are not given in the source, and metals are not pooled.
- Group tests treat strains as independent, although relatives share a^* (lambda 0.5-0.9). Their p-values are too small. Benjamini-Hochberg is applied across the 9 omnibus tests together.
- The dose-factor model (Table 2) and the per-group strata (Table 24) have no strain dose slope, so their intervals are optimistic.
- The regime split (section 6) is defined from colony size, which also relates to a^* .
- Size is partly an effect of stress. Models “at fixed size” estimate a direct effect, not the whole induction. One size slope is used for all strains.
- The area cutoff is justified for Cr and Cu from unstressed colonies and assumed for Pb. The Pb inhibitory range (doses 20-30) has too few colonies above any cutoff to test.
- Survivor selection: wells that never produced objects are absent, wells with fewer than 2 window images were dropped, and the area cutoff removes small colonies at the top doses. Top-dose estimates therefore describe larger colonies (Cr, Pb).
- 11 strains still have no species (“Species Not Found”, no genome) and are excluded from group tests. Groups need at least 5 strains. Pure haploid *R. mucilaginosa* has 172-173 strains and the others have 5-33.
- The sourmash species calls used in D-45 come from single 9003-batch libraries and are not verified.
- The hybrid group combines ploidy and parentage; its differences cannot be attributed to ploidy alone.
- The all-objects sensitivity table is built (`results/wells_allobj.csv`) but not modelled.
- The Cu b^* jump between dose 0 and dose 5 is not explained.

- No population-structure analysis is included until DH4148 SNP clusters exist (D-51).